

Math 521, Section 2
Homework 1
Selected Solutions

Problem 3. Let $f : A \rightarrow B$ be a function.

- (a) Prove that f is injective if and only if $f^{-1}(f(X)) = X$ for all $X \subseteq A$.
- (b) Prove that f is surjective if and only if $f(f^{-1}(Y)) = Y$ for all $Y \subseteq B$.

Solution: (b) $\boxed{\implies}$ Suppose that f is surjective, and fix $Y \subseteq B$. Recall from Lecture 2 that $f(f^{-1}(Y)) \subseteq Y$ for any function f , and so it only remains to show that $f(f^{-1}(Y)) \supseteq Y$. Fix $y \in Y$. As f is surjective, there exists $x \in X$ so that $f(x) = y$. Then $x \in f^{-1}(Y)$ by definition of preimage, and so $y = f(x) \in f(f^{-1}(Y))$ by definition of image.

$\boxed{\impliedby}$ Suppose that $f(f^{-1}(Y)) = Y$ for any $Y \subseteq B$. Fix $b \in B$. We want to show that there exists $a \in A$ so that $f(a) = b$. Consider the set $Y = \{b\}$. Then we know

$$f(f^{-1}(\{b\})) = \{b\}.$$

As $f(\emptyset) = \emptyset$, then we must have $f^{-1}(\{b\}) \neq \emptyset$. Therefore there exists a point $a \in f^{-1}(\{b\})$, and so $f(a) = b$ as desired. $\square$

Problem 5. Let $f : B \rightarrow C$ be a function. Prove that the following two statements are equivalent:

- (a) The function f is injective.
- (b) For any set A and for any functions $g : A \rightarrow B$ and $h : A \rightarrow B$ such that $f \circ g = f \circ h$, we have $g = h$.

Solution: (a) $\Rightarrow$ (b): Suppose that f is injective. We want to show that property (b) is true.

Let A be a set and $g, h : A \rightarrow B$ be functions such that $f \circ g = f \circ h$. This means that for any $a \in A$, we have

$$f(g(a)) = f(h(a)).$$

In other words, $g(a)$ and $h(a)$ are two points in B which f maps to the same point in C . As f is injective, we must have

$$g(a) = h(a).$$

This holds for any $a \in A$, and so the functions g and h are equal.

(a) $\Leftarrow$ (b): Suppose that f satisfies property (b). We want to show that f is injective. Fix $b_1, b_2 \in B$ such that $f(b_1) = f(b_2)$. Consider the functions $g, h : \{1\} \rightarrow B$ given by

$$g(1) = b_1 \quad \text{and} \quad h(1) = b_2.$$

Then we have $f \circ g = f \circ h$, since

$$f(g(1)) = f(b_1) = f(b_2) = f(h(1)).$$

Therefore, by property (b) we must have $g = h$, and so

$$b_1 = g(1) = h(1) = b_2.$$

Thus $b_1 = b_2$, as desired. □

Problem 6. Let $f : A \rightarrow B$ be a function. Suppose that there exists a function $g : B \rightarrow A$ such that $(g \circ f)(a) = a$ for all $a \in A$ and $(f \circ g)(b) = b$ for all $b \in B$. Prove that f is bijective.

Solution: First, we will show that f is injective. Suppose that $f(a_1) = f(a_2)$ for some $a_1, a_2 \in A$. Then

$$a_1 = g(f(a_1)) = g(f(a_2)) = a_2.$$

Next, we will show that f is surjective. Fix $b \in B$. Set $a = g(b)$. Then $a \in A$ and

$$f(a) = f(g(b)) = b.$$

Together, we conclude that f is bijective. □

Math 521, Section 2
Homework 2
Selected Solutions

Problem 1. Prove that if A and B are both countable sets and $A \cap B = \emptyset$, then $A \cup B$ is countable.

Idea: We'll use the same pattern that we used to prove that $\mathbb{Z}$ is countable in Lecture 3:

$$\begin{array}{ccccccccc} A \cup B : & \dots & b_2 & b_1 & a_1 & a_2 & a_3 & \dots \\ \mathbb{N} : & & 4 & 2 & 1 & 3 & 5 & \end{array}$$

Solution: As A and B are countable, there exist functions $g : A \rightarrow \mathbb{N}$ and $h : B \rightarrow \mathbb{N}$ that are bijective.

Define the function $f : A \cup B \rightarrow \mathbb{N}$ by

$$f(x) = \begin{cases} 2g(x) - 1 & x \in A, \\ 2h(x) & x \in B. \end{cases}$$

As $A \cap B = \emptyset$, then these two cases are mutually exclusive and so f is indeed a function.

We want to show that f is bijective. To this end, consider the function $\phi : \mathbb{N} \rightarrow A \cup B$ given by

$$\phi(n) = \begin{cases} g^{-1}(\frac{n+1}{2}) & n \text{ odd}, \\ h^{-1}(\frac{n}{2}) & n \text{ even}. \end{cases}$$

For $n \in \mathbb{N}$ odd, note that $\phi(n) = g^{-1}(\frac{n+1}{2}) \in A$ and so

$$(f \circ \phi)(n) = f(g^{-1}(\frac{n+1}{2})) = 2g(g^{-1}(\frac{n+1}{2})) - 1 = 2 \cdot \frac{n+1}{2} - 1 = n.$$

Similarly, for $n \in \mathbb{N}$ even, we have $\phi(n) = h^{-1}(\frac{n}{2}) \in B$ and so

$$(f \circ \phi)(n) = f(h^{-1}(\frac{n}{2})) = 2h(h^{-1}(\frac{n}{2})) = 2 \cdot \frac{n}{2} = n.$$

Together, we conclude that $(f \circ \phi)(n) = n$ for all $n \in \mathbb{N}$.

On the other hand, for $x \in A$ we observe that $f(x) = 2g(x) - 1$ is odd, and so

$$(\phi \circ f)(x) = \phi(2g(x) - 1) = g^{-1}(\frac{2g(x)-1+1}{2}) = g^{-1}(g(x)) = x.$$

Likewise, for $x \in B$ we see that $f(x) = 2h(x)$ is even, and so

$$(\phi \circ f)(x) = \phi(2h(x)) = h^{-1}(\frac{2h(x)}{2}) = h^{-1}(h(x)) = x.$$

Together, we conclude that $(\phi \circ f)(x) = x$ for all $x \in A \cup B$.

Combining the previous two paragraphs, we conclude that f is bijective by HW1#6. $\square$

Problem 3. Recall that given two sets A and B , we write $A \sim B$ if and only if there exists a bijective function $f : A \rightarrow B$. Prove that $\sim$ is an equivalence relation.

Solution: Reflexivity: Let A be a set. We want to show that $A \sim A$. Consider the function $f : A \rightarrow A$ given by $f(a) = a$. Notice that $(f \circ f)(a) = a$ for all $a \in A$, and so f is bijective by HW1#6. Therefore $A \sim A$.

Symmetry: Suppose that A, B are sets such that $A \sim B$. Then there exists a bijective function $g : A \rightarrow B$. Consider the function $g^{-1} : B \rightarrow A$. By Lecture 3 we know that $(g^{-1} \circ g)(a) = a$ for all $a \in A$ and $(g \circ g^{-1})(b) = b$ for all $b \in B$, and so g^{-1} is bijective by HW1#6. Therefore $B \sim A$.

Transitivity: Suppose that A, B, C are sets such that $A \sim B$ and $B \sim C$. Then there exist bijective functions $h_1 : A \rightarrow B$ and $h_2 : B \rightarrow C$. By Lecture 3, the composition $h_2 \circ h_1 : A \rightarrow C$ is also bijective, and so $A \sim C$. $\square$

Problem 5. Let $(F, +, \cdot)$ be a field.

- (a) Prove that for any $x \in F$, the function $f : F \rightarrow F$, $f(y) = x + y$ is bijective.
- (b) Prove that for any $x \in F \setminus \{0\}$, the function $g : F \rightarrow F$, $g(y) = x \cdot y$ is bijective.

Solution: (a) Fix $x \in F$. By (A5), there exists an additive inverse $-x \in F$ for x , so that

$$x + (-x) = 0 = -x + x.$$

Consider the function $h : F \rightarrow F$ given by $h(z) = -x + z$. Then, for any $z \in F$ we have

$$\begin{aligned} (f \circ h)(z) &= f(-x + z) = x + (-x + z) \\ &= (x + (-x)) + z && \text{by (A3)} \\ &= 0 + z && \text{by (A5)} \\ &= z && \text{by (A4)}. \end{aligned}$$

Similarly, for $y \in F$,

$$\begin{aligned} (h \circ f)(y) &= h(x + y) = -x + (x + y) \\ &= (-x + x) + y && \text{by (A3)} \\ &= 0 + y && \text{by (A5)} \\ &= y && \text{by (A4)}. \end{aligned}$$

Therefore f is bijective by HW1#6.

- (b) Fix $x \in F \setminus \{0\}$. As $x \neq 0$, there exists a multiplicative inverse $x^{-1} \in F$ for x , so that

$$x \cdot x^{-1} = 1 = x^{-1} \cdot x.$$

Consider the function $h : F \rightarrow F$ given by $h(z) = x^{-1} \cdot z$. Then, for any $z \in F$ we have

$$\begin{aligned} (g \circ h)(z) &= g(x^{-1} \cdot z) = x \cdot (x^{-1} \cdot z) \\ &= (x \cdot x^{-1}) \cdot z && \text{by (M3)} \\ &= 1 \cdot z && \text{by (M5)} \\ &= z && \text{by (M4)}. \end{aligned}$$

Similarly, for $y \in F$,

$$\begin{aligned} (h \circ g)(y) &= h(x \cdot y) = x^{-1} \cdot (x \cdot y) \\ &= (x^{-1} \cdot x) \cdot y && \text{by (M3)} \\ &= 1 \cdot y && \text{by (M5)} \\ &= y && \text{by (M4)}. \end{aligned}$$

Therefore g is bijective by HW1#6. □

Problem 6. Let $(F, +, \cdot)$ be a field. Suppose that $F = \{0, 1, a, b\}$ is a set with exactly four elements, where 0 and 1 denote the identities for $+$ and $\cdot$ respectively, and a, b denote the remaining two elements of F . Fill in the addition and multiplication tables below, and justify your answer.

$+$	0	1	a	b
0				
1				
a				
b				

$\cdot$	0	1	a	b
0				
1				
a				
b				

Solution: (There are many ways to do this problem. Here's one option that works.)

Step 1: Let's start with the multiplication table. We can fill in the 0 and 1 row/column using what we know from Lecture 5:

$\cdot$	0	1	a	b
0	0	0	0	0
1	0	1	a	b
a	0	a		
b	0	b		

Step 2: Notice that the only possibility for the $a \cdot b$ entry is 1, since the other three elements already appear in this row and column. This in turn leaves us with only one option for the remaining entries:

$\cdot$	0	1	a	b
0	0	0	0	0
1	0	1	a	b
a	0	a	b	1
b	0	b	1	a

Step 3: Now let's turn to the addition table. We can fill in the 0 row/column using (A4):

$+$	0	1	a	b
0	0	1	a	b
1	1			
a	a			
b	b			

Step 4: Consider the entry for $1 + a$. There are two elements that we can choose from, 0 and b , since the other two elements already appear in this row and column.

Suppose $1 + a = 0$. This forces

$$\begin{aligned}
 0 &= 0 \cdot b = (1 + a) \cdot b \\
 &= 1 \cdot b + a \cdot b \quad \text{by (D)} \\
 &= b + 1 \quad \text{by the multiplication table.}
 \end{aligned}$$

But this would mean that 0 appears twice in a row/column, which is a contradiction:

+	0	1	a	b
0	0	1	a	b
1	1		0	0
a	a	0		
b	b	0		

Therefore we must have $1 + a = b$. Entering this into our table, we can then fill in the entries

+	0	1	a	b
0	0	1	a	b
1	1		b	
a	a	b		
b	b			

Similarly, if we had $1 + b = 0$, then

$$\begin{aligned}
0 &= 0 \cdot b = (1 + b) \cdot a \\
&= 1 \cdot a + b \cdot a \quad \text{by (D)} \\
&= a + 1 \quad \text{by the multiplication table,}
\end{aligned}$$

which would contradict our current entry for $a + 1$. So we must have $1 + b = a$. Filling this in, we can complete the second row/column:

+	0	1	a	b
0	0	1	a	b
1	1	0	b	a
a	a	b		
b	b	a		

Step 5: Consider the entry for $a + b$. There are two elements that we can choose from, 0 and 1, since the other two elements already appear in this row and column.

Suppose $a + b = 0$. This forces

$$\begin{aligned}
0 &= 0 \cdot b = (a + b) \cdot b \\
&= a \cdot b + b \cdot b \quad \text{by (D)} \\
&= 1 + a \quad \text{by the multiplication table.}
\end{aligned}$$

This contradicts our current entry for $1 + a$, so we conclude that $a + b = 1$. This in turn leaves us with only one option for the remaining entries:

+	0	1	a	b
0	0	1	a	b
1	1	0	b	a
a	a	b	0	1
b	b	a	1	0

□

Math 521, Section 2
Homework 3
Solutions

Problem 1. Let $(F, +, \cdot)$ be a field.

- (a) Prove that the multiplicative identity is unique: If $b \in F$ satisfies $b \cdot a = a \cdot b = a$ for all $a \in F$, then $b = 1$.
- (b) Prove that multiplicative inverses are unique: If $a \in F \setminus \{0\}$ and $b \in F$ satisfy $b \cdot a = a \cdot b = 1$, then $b = a^{-1}$.
- (c) Prove that if $a \cdot b = a \cdot c$ and $a \neq 0$, then $b = c$.
- (d) Prove that if $a, b \in F$ then $-a + (-b) = -(a + b)$.
- (e) Prove that if $a, b \in F$ and $a, b \neq 0$, then $a^{-1} \cdot b^{-1} = (ab)^{-1}$.

Solution: (a) We know that $1, b \in F$ satisfy:

- $a \cdot 1 = 1 \cdot a = a$ for all $a \in A$
- $a \cdot b = b \cdot a = a$ for all $a \in A$

Taking $a = b$ in the first bullet point, we see that

$$b \cdot 1 = b.$$

Taking $a = 1$ in the second bullet point, we obtain

$$b \cdot 1 = 1.$$

Comparing the previous two equalities, we conclude $b = 1$.

- (b) As $a \neq 0$, then by M5 we know there exists $a^{-1} \in F$ so that

$$a \cdot a^{-1} = a^{-1} \cdot a = 1.$$

Multiplying $b \cdot a = 1$ by a^{-1} on the right, we obtain

$$\begin{aligned} (b \cdot a) \cdot a^{-1} &= 1 \cdot a^{-1} \implies b \cdot (a \cdot a^{-1}) = a^{-1} && \text{by M3 and M4} \\ &\implies b \cdot 1 = a^{-1} && \text{by M5} \\ &\implies b = a^{-1} && \text{by M4.} \end{aligned}$$

- (c) As $a \neq 0$, then by M5 we know there exists $a^{-1} \in F$ so that

$$a \cdot a^{-1} = a^{-1} \cdot a = 1.$$

Multiplying $a \cdot b = a \cdot c$ by a^{-1} on the left, we obtain

$$\begin{aligned} a^{-1} \cdot (a \cdot b) &= a^{-1} \cdot (a \cdot c) \implies (a^{-1} \cdot a) \cdot b = (a^{-1} \cdot a) \cdot c && \text{by M3} \\ &\implies 1 \cdot b = 1 \cdot c && \text{by M5} \\ &\implies b = c && \text{by M4.} \end{aligned}$$

(d) Fix $a, b \in F$. Then

$$\begin{aligned}(-a + (-b)) + (a + b) &= (-a + (-b)) + (b + a) && \text{by (A2)} \\&= -a + ((-b) + b) + a && \text{by (A3)} \\&= -a + 0 + a && \text{by (A5)} \\&= -a + a && \text{by (A4)} \\&= 0 && \text{by (A5)}.\end{aligned}$$

In other words, $-a + (-b)$ is the additive inverse of $a + b$. As additive inverses are unique, then we conclude $-a + (-b) = -(a + b)$.

(e) Fix $a, b \neq 0$. Then

$$\begin{aligned}(a^{-1} \cdot b^{-1}) \cdot (ab) &= (a^{-1} \cdot b^{-1}) \cdot (ba) && \text{by (M2)} \\&= a^{-1} \cdot (b^{-1}b) \cdot a && \text{by (M3)} \\&= a^{-1} \cdot 1 \cdot a && \text{by (M5)} \\&= a^{-1} \cdot a && \text{by (M4)} \\&= 1 && \text{by (M5)}.\end{aligned}$$

In other words, $a^{-1} \cdot b^{-1}$ is the multiplicative inverse of ab . As multiplicative inverses are unique (by part (b)), then we conclude $a^{-1} \cdot b^{-1} = (ab)^{-1}$. $\square$

Problem 2. Given two points (x_1, y_1) and (x_2, y_2) in $\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$, we write

$$(x_1, y_1) < (x_2, y_2) \iff x_1 < x_2 \text{ or } (x_1 = x_2 \text{ and } y_1 < y_2).$$

- (a) Prove that $<$ is an order relation on $\mathbb{R}^2$.
- (b) Prove that $(\mathbb{R}^2, <)$ does not satisfy the least upper bound property.

Solution: (a) By definition of an order relation, there are two properties to check.

Trichotomy: Fix $(x, y), (x', y') \in \mathbb{R}^2$. We will prove that in every possible case, exactly one of the statements ' $<$ ', ' $=$ ', or ' $>$ ' is true.

- Case: $x < x'$. Then $(x, y) < (x', y')$.
- Case: $x > x'$. Then $(x, y) > (x', y')$.
- Case: $x = x'$.
 - If $y < y'$: Then $(x, y) < (x', y')$.
 - If $y > y'$: Then $(x, y) > (x', y')$.
 - If $y = y'$: Then $(x, y) = (x', y')$.

Transitivity: Suppose $(x_1, y_1) < (x_2, y_2)$ and $(x_2, y_2) < (x_3, y_3)$. By definition of ' $<$ ', there are four possible cases.

- Case: $x_1 < x_2$ and $x_2 < x_3$. Then $x_1 < x_3$, and so $(x_1, y_1) < (x_3, y_3)$.
- Case: $x_1 < x_2$, $x_2 = x_3$, and $y_2 < y_3$. Then $x_1 < x_3$, and so $(x_1, y_1) < (x_3, y_3)$.
- Case: $x_1 = x_2$, $y_1 < y_2$, and $x_2 < x_3$. Then $x_1 < x_3$, and so $(x_1, y_1) < (x_3, y_3)$.
- Case: $x_1 = x_2$, $y_1 < y_2$, $x_2 = x_3$, and $y_2 < y_3$. Then $x_1 = x_3$ and $y_1 < y_3$, and so $(x_1, y_1) < (x_3, y_3)$.

So, in all cases, we have $(x_1, y_1) < (x_3, y_3)$.

(b) Suppose towards a contradiction that $(\mathbb{R}^2, <)$ satisfies the least upper bound property. Set $Y = \{(0, y) : y \in \mathbb{R}\}$. Then:

- Y is nonempty: since $(0, 0) \in Y$.
- Y is bounded above: by $(1, 0)$.

So, by the least upper bound property, there exists a least upper bound $(a, b) \in \mathbb{R}^2$ for Y .

Claim: $a > 0$. If $a < 0$, then $(0, 0)$ would be a point in Y with $(0, 0) > (a, b)$, which would contradict that (a, b) is an upper bound for Y . If $a = 0$, then $(0, b + 1)$ would be a point in Y with $(0, b + 1) > (a, b)$, which would contradict that (a, b) is an upper bound for Y . The only remaining possibility is $a > 0$.

As $a > 0$, then $(\frac{a}{2}, b)$ is also an upper bound for Y :

$$0 < \frac{a}{2} \implies (0, y) < (\frac{a}{2}, b) \quad \forall y \in \mathbb{R}.$$

On the other hand, we have $\frac{a}{2} < a$, and so $(\frac{a}{2}, b) < (a, b)$. This contradicts that (a, b) was the least upper bound. $\square$

Problem 3. Recall that the set of complex numbers $\mathbb{C}$ consists of all numbers of the form $a + ib$ for $a, b \in \mathbb{R}$. We can add and multiply complex numbers as follows:

$$(a + ib) + (c + id) = (a + c) + i(b + d) \quad \text{and} \quad (a + ib) \cdot (c + id) = (ac - bd) + i(ad + bc).$$

With these operations, it turns out $(\mathbb{C}, +, \cdot)$ is a field (but you do not need to prove this).

- (a) The definition of a field says that for any $a + ib \neq 0 + i0$, there exists $c + id \in \mathbb{C}$ so that $(a + ib) \cdot (c + id) = 1 + i0$. What are $c, d \in \mathbb{R}$ in terms of $a, b \in \mathbb{R}$? Prove your answer.
- (b) Prove that there is no order relation $<$ on $\mathbb{C}$ that makes $(\mathbb{C}, +, \cdot, <)$ an ordered field.

Scratch work:

$$\frac{1}{a + ib} = \frac{a - ib}{(a + ib)(a - ib)} = \frac{a - ib}{a^2 + b^2} = \frac{a}{a^2 + b^2} - i \frac{b}{a^2 + b^2}$$

Solution: (a) Set

$$c = \frac{a}{a^2 + b^2}, \quad d = -\frac{b}{a^2 + b^2}.$$

Then

$$\begin{aligned} ac - bd &= \frac{a^2}{a^2 + b^2} + \frac{b^2}{a^2 + b^2} = 1, \\ ad + bc &= -\frac{ab}{a^2 + b^2} + \frac{ab}{a^2 + b^2} = 0, \end{aligned}$$

and so $(a + ib) \cdot (c + id) = 1$.

(b) Suppose towards a contradiction that $(\mathbb{C}, +, \cdot, <)$ is an ordered field. As $i \neq 0$, then by a proposition from Lecture 7 we know that

$$0 < i^2 = -1.$$

On the other hand, we also proved in Lecture 5 (in the same proposition) that

$$1 > 0, \quad \text{and so} \quad -1 < 0.$$

Altogether we have both $-1 > 0$ and $-1 < 0$, which is a contradiction. □

Problem 4. Let $A \subseteq \mathbb{R}$ be a nonempty and bounded set, let $c \in \mathbb{R}$ be a number, and define

$$B = \{c \cdot a : a \in A\}.$$

- (a) Prove that if $c > 0$, then $\sup B = c \cdot \sup A$.
- (b) Prove that if $c < 0$, then $\sup B = c \cdot \inf A$.

Solution: (a) As A is nonempty and bounded, then $\sup A \in \mathbb{R}$ exists.

Claim: $c \sup A$ is an upper bound for B . As $\sup A$ is an upper bound for A , we know that

$$a \leq \sup A \quad \text{for all } a \in A,$$

and so

$$ca \leq c \sup A \quad \text{for all } a \in A.$$

Therefore $c \sup A$ is an upper bound for B .

Claim: If L is an upper bound for B , then $L \geq c \sup A$. As L is an upper bound for B , we have

$$ca \leq L \quad \text{for all } a \in A,$$

and so

$$a \leq \frac{L}{c} \quad \text{for all } a \in A.$$

In other words, $\frac{L}{c}$ is an upper bound for A . As $\sup A$ is the least upper bound, we must have

$$\sup A \leq \frac{L}{c},$$

and therefore

$$c \sup A \leq L$$

as desired.

Collecting the previous two claims, we conclude that $c \sup A = \sup B$ by definition of $\sup B$.

- (b) As A is nonempty and bounded, then $\inf A \in \mathbb{R}$ exists.

Claim: $c \inf A$ is an upper bound for B . As $\inf A$ is a lower bound for A , we know that

$$a \geq \inf A \quad \text{for all } a \in A,$$

and so

$$ca \leq c \inf A \quad \text{for all } a \in A.$$

Therefore $c \inf A$ is an upper bound for B .

Claim: If L is an upper bound for B , then $L \geq c \inf A$. As L is an upper bound for B , we have

$$ca \leq L \quad \text{for all } a \in A,$$

and so

$$a \geq \frac{L}{c} \quad \text{for all } a \in A.$$

In other words, $\frac{L}{c}$ is a lower bound for A . As $\inf A$ is the greatest lower bound, we must have

$$\inf A \geq \frac{L}{c},$$

and therefore

$$c \inf A \leq L$$

as desired.

Collecting the previous two claims, we conclude that $c \inf A = \sup B$ by definition of $\sup B$. $\square$

Problem 5. Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets, and define

$$S = \{a + b : a \in A \text{ and } b \in B\}.$$

Prove that $\sup S = \sup A + \sup B$.

Solution: As $A, B \subseteq \mathbb{R}$ are nonempty and bounded, we know that $\sup A$ and $\sup B$ exist. We will prove that $\sup A + \sup B$ satisfies the definition of ‘least upper bound’ for the set S .

Claim: $\sup A + \sup B$ is an upper bound for S . As $\sup A$ is an upper bound for A , we know

$$a \leq \sup A \quad \text{for all } a \in A.$$

Similarly, $\sup B$ is an upper bound for B , and so

$$b \leq \sup B \quad \text{for all } b \in B.$$

Adding these two inequalities, we conclude that

$$a + b \leq \sup A + \sup B \quad \text{for all } a \in A \text{ and } b \in B.$$

Therefore $\sup A + \sup B$ is an upper bound for S .

Claim: If L is an upper bound for S , then $L \geq \sup A + \sup B$. Suppose towards a contradiction that L is an upper bound for S but $L < \sup A + \sup B$. Set

$$\epsilon = \frac{1}{2}(\sup A + \sup B - L) > 0.$$

As $\sup A - \epsilon < \sup A$, then $\sup A - \epsilon$ cannot be an upper bound for A , and so there exists $a_0 \in A$ so that

$$a_0 > \sup A - \epsilon.$$

Similarly, as $\sup B - \epsilon$ is not an upper bound for B , then there exists $b_0 \in B$ so that

$$b_0 > \sup B - \epsilon.$$

Adding these two inequalities and using the definition of ϵ , we deduce

$$a_0 + b_0 > \sup A + \sup B - 2\epsilon = L.$$

This contradicts that L is an upper bound for S , and so we must have $L \geq \sup A + \sup B$. $\square$

Problem 6. Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets. Prove or disprove the following statements:

- (a) If $a \leq b$ for all $a \in A$ and for all $b \in B$, then $\sup A \leq \inf B$.
- (b) If $a < b$ for all $a \in A$ and for all $b \in B$, then $\sup A < \inf B$.

Solution: (a) **Prove:** As $A, B \subseteq \mathbb{R}$ are nonempty and bounded, then both $\sup A, \inf B \in \mathbb{R}$ exist.

Let $b \in B$. As $a \leq b$ for all $a \in A$, then b is an upper bound for A , and so

$$\sup A \leq b.$$

This is true for all $b \in B$. Therefore $\sup A$ is a lower bound for B , and so $\sup A \leq \inf B$.

- (b) (Recall that in order to disprove a statement, we must provide a counterexample.)

Disprove: Consider the sets

$$A = \{1 - \frac{1}{n} : n \in \mathbb{N}\}, \quad B = \{1\}.$$

Note that $\inf B = \min B = 1$ since B is a finite set, and recall (from Lecture 7) that $\sup A = 1$.

Altogether:

- A, B are nonempty
- A, B are bounded: 0 is a lower bound and 1 is an upper bound, for both A and B .
- $a < b$ for all $a \in A$ and $b \in B$: since $1 - \frac{1}{n} < 1$ for all $n \geq 1$.
- $\sup A \not< \inf B$: since $\sup A = 1 = \inf B$.

□

Math 521, Section 2
Homework 4
Solutions

Problem 1. Let $\{a_n\}_{n \geq 1}$ be a sequence.

- (a) Prove that if $\{a_n\}_{n \geq 1}$ converges to a , then the sequence $\{|a_n|\}_{n \geq 1}$ converges to $|a|$.
- (b) Provide an example where $\{|a_n|\}_{n \geq 1}$ converges, but $\{a_n\}_{n \geq 1}$ diverges.

Solution: (a) Fix $\epsilon > 0$. As $\lim_{n \rightarrow \infty} a_n = a$, there exists $N \in \mathbb{N}$ so that

$$|a_n - a| < \epsilon \quad \forall n \geq N.$$

By the reverse triangle inequality, we have

$$||a_n| - |a|| \leq |a_n - a| < \epsilon \quad \forall n \geq N.$$

So $\lim_{n \rightarrow \infty} |a_n| = |a|$.

(b) By Lecture 10, the sequence $\{a_n\}_{n \geq 1}$ where $a_n = (-1)^n$ diverges. On the other hand, we have $|a_n| = 1$ for all $n \geq 1$, and so the sequence $\{|a_n|\}_{n \geq 1}$ converges to 1. $\square$

Problem 2 (Squeeze theorem). Prove that if $\{a_n\}_{n \geq 1}$, $\{b_n\}_{n \geq 1}$, and $\{c_n\}_{n \geq 1}$ are sequences such that

$$a_n \leq b_n \leq c_n \text{ for all } n \geq 1 \quad \text{and} \quad \lim_{n \rightarrow \infty} a_n = L = \lim_{n \rightarrow \infty} c_n,$$

then $\{b_n\}_{n \geq 1}$ converges and $\lim_{n \rightarrow \infty} b_n = L$.

Solution: Fix $\epsilon > 0$. As $\lim_{n \rightarrow \infty} a_n = L$, there exists $N_1 \in \mathbb{N}$ so that

$$|a_n - L| < \epsilon \quad \forall n \geq N_1.$$

Similarly, as $\lim_{n \rightarrow \infty} c_n = L$, there exists $N_2 \in \mathbb{N}$ so that

$$|c_n - L| < \epsilon \quad \forall n \geq N_2.$$

Set $N = \max\{N_1, N_2\}$. Then, for $n \geq N$ we have

$$b_n - L \leq c_n - L < \epsilon \quad \text{and} \quad b_n - L \geq a_n - L > -\epsilon.$$

Together, we conclude

$$|b_n - L| < \epsilon \quad \forall n \geq N.$$

So $\lim_{n \rightarrow \infty} b_n = L$, as desired. □

Problem 3. Suppose that the sequence $\{a_n\}_{n \geq 1}$ converges.

- (a) Prove that if $a_n \geq m$ for all but finitely many n , then $\lim_{n \rightarrow \infty} a_n \geq m$.
(b) Prove that if $a_n \leq M$ for all but finitely many n , then $\lim_{n \rightarrow \infty} a_n \leq M$.

Solution: (a) Set $a = \lim_{n \rightarrow \infty} a_n$. Let $N_1 \in \mathbb{N}$ be larger than the largest index n for which $a_n \not\geq m$. Then

$$a_n \geq m \quad \forall n \geq N_1.$$

Fix $\epsilon > 0$. As $\lim_{n \rightarrow \infty} a_n = a$, there exists $N_2 \in \mathbb{N}$ so that

$$|a_n - a| < \epsilon \quad \forall n \geq N_2.$$

Set $N = \max\{N_1, N_2\}$. Then $N \geq N_1$ and $N \geq N_2$, and so

$$a - m = (a - a_N) + (a_N - m) > -\epsilon + 0.$$

As $\epsilon > 0$ was arbitrary, we must have $a - m \geq 0$. (Otherwise, we would have $a - m < 0$, and then picking $\epsilon = m - a > 0$ would yield a contradiction.)

(b) Set $a = \lim_{n \rightarrow \infty} a_n$. Let $N_1 \in \mathbb{N}$ be larger than the largest index n for which $a_n \not\leq M$. Then

$$a_n \leq M \quad \forall n \geq N_1.$$

Fix $\epsilon > 0$. As $\lim_{n \rightarrow \infty} a_n = a$, there exists $N_2 \in \mathbb{N}$ so that

$$|a_n - a| < \epsilon \quad \forall n \geq N_2.$$

Set $N = \max\{N_1, N_2\}$. Then

$$a - M = (a - a_N) + (a_N - M) < \epsilon + 0.$$

As $\epsilon > 0$ was arbitrary, we must have $a - M \leq 0$. □

Problem 4. Suppose that the sequence $\{a_n\}_{n \geq 1}$ converges. Prove or disprove the following statements:

- (a) If $\lim_{n \rightarrow \infty} a_n \geq m$, then there exists $N \in \mathbb{N}$ so that $a_n \geq m$ for all $n \geq N$.
- (b) If $\lim_{n \rightarrow \infty} a_n > m$, then there exists $N \in \mathbb{N}$ so that $a_n > m$ for all $n \geq N$.

Solution: (a) **Disprove.** Consider the sequence $\{a_n\}_{n \geq 1}$ with $a_n = -\frac{1}{n}$. From Lecture 10 we know that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, and so $a = \lim_{n \rightarrow \infty} a_n = 0$ as well by the limit law for multiplication by a constant. So $m = 0$ is a number that satisfies

$$a = 0 \geq 0 = m.$$

On the other hand, we have

$$a_n = -\frac{1}{n} \not\geq 0 = m$$

for all $n \in \mathbb{N}$, and so no such N can exist.

(b) **Prove.** Suppose $a > m$ and set $\epsilon = a - m$. Then $\epsilon > 0$, and so there exists $N \in \mathbb{N}$ so that

$$|a_n - a| < \epsilon \quad \forall n \geq N.$$

Then, for $n \geq N$ we have

$$a_n - m = (a_n - a) + (a - m) > -\epsilon + (a - m) = 0.$$

Therefore $a_n > m$ for all $n \geq N$, as desired. □

Problem 5. Prove that if $\{a_n\}_{n \geq 1}$ is a sequence of positive numbers that converges to $a > 0$, then the sequence $\{\sqrt{a_n}\}_{n \geq 1}$ converges to $\sqrt{a}$.

Solution: Fix $\epsilon > 0$. As $\lim_{n \rightarrow \infty} a_n = a$ and $a > 0$, there exists $N \in \mathbb{N}$ so that

$$|a_n - a| < \sqrt{a}\epsilon \quad \forall n \geq N.$$

Then, for $n \geq N$ we have

$$|\sqrt{a_n} - \sqrt{a}| = \left| \frac{(\sqrt{a_n} - \sqrt{a})(\sqrt{a_n} + \sqrt{a})}{\sqrt{a_n} + \sqrt{a}} \right| = \frac{|a_n - a|}{\sqrt{a_n} + \sqrt{a}} \leq \frac{|a_n - a|}{\sqrt{a}} < \frac{\sqrt{a}\epsilon}{\sqrt{a}} = \epsilon.$$

(In the third step above, we noted that $\sqrt{a_n} + \sqrt{a} \geq \sqrt{a}$ since $a_n > 0$.) Therefore $\lim_{n \rightarrow \infty} \sqrt{a_n} = \sqrt{a}$, as desired. $\square$

Problem 6. Consider the following sequence:

$$a_1 = \sqrt{2} \quad \text{and} \quad a_{n+1} = \sqrt{2 + a_n} \quad \text{for all } n \geq 1.$$

- (a) Prove that the sequence $\{a_n\}_{n \geq 1}$ is bounded.
- (b) Prove that the sequence is increasing.
- (c) Conclude that the sequence converges, and find its limit.

Solution: (a) We will prove $\sqrt{2} \leq a_n < 2$ for all $n \in \mathbb{N}$ by induction.

Base case: For $n = 1$, the statement $\sqrt{2} \leq \sqrt{2} < 2$ is true.

Inductive step: Suppose that $\sqrt{2} \leq a_n < 2$ for some $n \in \mathbb{N}$. Then

$$a_{n+1} = \sqrt{2 + a_n} > \sqrt{2 + 0} = \sqrt{2},$$

$$a_{n+1} = \sqrt{2 + a_n} < \sqrt{2 + 2} = 2.$$

Together we obtain $\sqrt{2} \leq a_{n+1} < 2$.

By induction, we conclude that $\sqrt{2} \leq a_n < 2$ holds for all $n \in \mathbb{N}$.

(b) Fix $n \in \mathbb{N}$. Then

$$\begin{aligned} a_n \leq a_{n+1} = \sqrt{2 + a_n} &\iff a_n^2 \leq 2 + a_n \quad (\text{since } a_n > 0 \text{ by (a)}) \\ &\iff 0 \geq a_n^2 - a_n - 2 = (a_n - 2)(a_n + 1) \end{aligned}$$

By part (a), we know that $a_n - 2 < 0$ and $a_n + 1 > 0$, and so $(a_n - 2)(a_n + 1) < 0$. Working backwards through the above implications, we conclude that $a_n \leq a_{n+1}$.

(c) As the sequence $\{a_n\}_{n \geq 1}$ is increasing and bounded, then it converges by Lecture 11. By part (a) and problem 3,

$$a = \lim_{n \rightarrow \infty} a_n \quad \text{satisfies} \quad \sqrt{2} \leq a \leq 2.$$

By the limit law for addition and problem 5, we have

$$\lim_{n \rightarrow \infty} (2 + a_n) = 2 + a > 0, \quad \text{and so} \quad \lim_{n \rightarrow \infty} \sqrt{2 + a_n} = \sqrt{2 + a}.$$

So, taking the limit $n \rightarrow \infty$ in

$$a_{n+1} = \sqrt{2 + a_n},$$

we see that a satisfies

$$\begin{aligned} a = \sqrt{2 + a} &\implies a^2 = 2 + a \\ &\implies 0 = a^2 - a - 2 = (a - 2)(a + 1) \\ &\implies a = 2 \text{ or } a = -1. \end{aligned}$$

The case $a = -1$ is not possible, since we know $a \geq \sqrt{2}$. Therefore $a = 2$. □

Math 521, Section 2
Homework 5
Selected Solutions

Problem 1. Let $\{a_n\}_{n \geq 1}$ be a sequence of positive real numbers. Prove that $\lim_{n \rightarrow \infty} a_n = +\infty$ if and only if $\lim_{n \rightarrow \infty} \frac{1}{a_n} = 0$.

Solution: $\Rightarrow$ Fix $\epsilon > 0$. As $\lim a_n = +\infty$, then there exists N so that

$$a_n > \frac{1}{\epsilon} \quad \text{for all } n \geq N.$$

So, for $n \geq N$ we have

$$-\epsilon < 0 < \frac{1}{a_n} < \epsilon \quad \implies \quad \left| \frac{1}{a_n} - 0 \right| < \epsilon.$$

Therefore $\lim_{n \rightarrow \infty} \frac{1}{a_n} = 0$.

$\Leftarrow$ Fix $M > 0$. As $\lim_{n \rightarrow \infty} \frac{1}{a_n} = 0$, then there exists N so that

$$\left| \frac{1}{a_n} - 0 \right| < \frac{1}{M} \quad \text{for all } n \geq N.$$

Note that $\left| \frac{1}{a_n} - 0 \right| = \frac{1}{a_n} > 0$. So

$$a_n > M \quad \text{for all } n \geq N.$$

Therefore $\lim a_n = +\infty$. □

Problem 2. Let $\{a_n\}_{n \geq 1}$ be a sequence. Prove that if $\{a_{k_n}\}_{n \geq 1}$ is a subsequence of $\{a_n\}_{n \geq 1}$ that converges to $L \in \mathbb{R}$, then

$$\liminf_{n \rightarrow \infty} a_n \leq L \leq \limsup_{n \rightarrow \infty} a_n.$$

Solution: Let's start with the second inequality:

$$L \leq \limsup_{n \rightarrow \infty} a_n.$$

Case: $\limsup a_n = +\infty$. Then the statement is automatically true, since every real number L satisfies $L < +\infty$.

Case: $\limsup a_n = -\infty$. As $\liminf a_n \leq \limsup a_n$, then we must have $\liminf a_n = -\infty = \limsup a_n$. This means that $\lim a_n = -\infty$ by Lecture 14.

Fix $m < 0$. As $\lim a_n = -\infty$, there exists N so that

$$a_n < m \quad \text{for all } n \geq N.$$

As $k_n \geq n$ for all n , this implies

$$a_{k_n} < m \quad \text{for all } n \geq N.$$

Therefore $L = \lim a_{k_n} = -\infty$. This contradicts that $L \in \mathbb{R}$, and so this case is impossible.

Case: $\limsup a_n \in \mathbb{R}$. For $N \geq 1$, notice that

$$a_{k_n} \leq \sup\{a_n : n \geq N\} \quad \text{for all } n \geq N.$$

We know that the sequence $\{a_{k_n}\}$ converges to L . So, by HW4#3, we have

$$L = \lim a_{k_n} \leq \sup\{a_n : n \geq N\}.$$

On the other hand, $v_N = \sup\{a_n : n \geq N\}$ is a sequence that converges to $\limsup a_n$. So, using HW4#3 again, we conclude that

$$L \leq \lim_{N \rightarrow \infty} \sup\{a_n : n \geq N\} = \limsup a_n.$$

Exercise: Using similar steps, prove that

$$\liminf_{n \rightarrow \infty} a_n \leq L.$$

□

Problem 3. Let $\{a_n\}_{n \geq 1}$ and $\{b_n\}_{n \geq 1}$ be two bounded sequences. Prove that

$$\limsup_{n \rightarrow \infty} (a_n + b_n) \leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n.$$

Solution: Fix $N \geq 1$. For $n \geq N$ we have

$$a_n \leq \sup\{a_n : n \geq N\}, \quad b_n \leq \sup\{b_n : n \geq N\},$$

and so

$$a_n + b_n \leq \sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}.$$

In other words, $\sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}$ is an upper bound for the set $\{a_n + b_n : n \geq N\}$. Therefore

$$\sup\{a_n + b_n : n \geq N\} \leq \sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}.$$

Altogether, we have shown that for any $N \geq 1$ we have

$$\sup\{a_n + b_n : n \geq N\} \leq \sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}.$$

As $\{a_n\}$ is a bounded sequence, we know from lecture that the increasing sequence $v_N = \sup\{a_n : n \geq N\}$ converges to $v = \limsup a_n \in \mathbb{R}$. Similarly, as $\{b_n\}$ is bounded, the sequence $\sup\{b_n : n \geq N\}$ converges to $\limsup b_n$ as $N \rightarrow \infty$. By the inequality above, we see that the sequence $\sup\{a_n + b_n : n \geq N\}$ is also bounded above. Therefore $\limsup(a_n + b_n)$ is a finite number, and $\sup\{a_n + b_n : n \geq N\}$ converges to $\limsup(a_n + b_n)$ as $N \rightarrow \infty$.

By the limit laws, the sequence

$$\sup\{a_n + b_n : n \geq N\} - \sup\{a_n : n \geq N\} - \sup\{b_n : n \geq N\}$$

converges as $N \rightarrow \infty$. We also know that it is ≤ 0 for all $N \geq 1$. Therefore, by HW4#3, the limit is also ≤ 0 :

$$\begin{aligned} \limsup_{n \rightarrow \infty} (a_n + b_n) - \limsup_{n \rightarrow \infty} a_n - \limsup_{n \rightarrow \infty} b_n &\leq 0 \\ \implies \limsup_{n \rightarrow \infty} (a_n + b_n) &\leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n. \end{aligned}$$

□

Problem 4. Let $\{a_n\}_{n \geq 1}$ be a sequence. Prove that $\limsup_{n \rightarrow \infty} |a_n| = 0$ if and only if $\lim_{n \rightarrow \infty} a_n = 0$.

Solution: $\Rightarrow$ Suppose $\limsup |a_n| = 0$.

Claim: $\liminf |a_n| \geq 0$. As $|a_n| \geq 0$ for all n , we have

$$\inf_{n \geq N} |a_n| \geq 0 \quad \text{for all } N \geq 1.$$

Therefore

$$\lim_{N \rightarrow \infty} \inf_{n \geq N} |a_n| \geq 0$$

by HW4#3.

Altogether, we have

$$0 \leq \liminf |a_n| \leq \limsup |a_n| = 0.$$

Therefore

$$\liminf |a_n| = 0 = \limsup |a_n|.$$

By Lecture 10, this means that

$$\lim |a_n| = 0.$$

In other words: for all $\epsilon > 0$ there exists N so that

$$||a_n| - 0| < \epsilon \quad \text{for all } n \geq N.$$

This is the same as: for all $\epsilon > 0$ there exists N so that

$$|a_n - 0| < \epsilon \quad \text{for all } n \geq N.$$

So $\lim a_n = 0$.

$\Leftarrow$ Suppose $\lim a_n = 0$. By HW4#1, this implies

$$\lim |a_n| = 0.$$

By Lecture 14, we have

$$\liminf |a_n| = 0 = \limsup |a_n|.$$

So $\limsup |a_n| = 0$, as desired. □

Math 521, Section 2
Homework 6
Selected Solutions

Problem 2. Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be convergent series.

(a) Prove that for any $k \in \mathbb{R}$, the series $\sum_{n=1}^{\infty} ka_n$ converges and

$$\sum_{n=1}^{\infty} ka_n = k \sum_{n=1}^{\infty} a_n.$$

(b) Prove that the series $\sum_{n=1}^{\infty} (a_n + b_n)$ converges and

$$\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n.$$

Solution: (a) Let $s_n = \sum_{j=1}^n a_j$ denote the n th partial sum. As $\sum a_n$ converges, we know that the sequence $\{s_n\}$ converges.

Fix $k \in \mathbb{R}$. Then the partial sums $\sum_{j=1}^n ka_j$ are equal to ks_n . By the limit law for multiplication by a constant, we conclude that the sequence $\{ks_n\}_{n \geq 1}$ converges and $\lim(ks_n) = k \lim s_n$. By definition, this is exactly what it means for the series $\sum ka_n$ to converge to the number $k \sum a_n$.

(b) Let $s_n = \sum_{j=1}^n a_j$ and $t_n = \sum_{j=1}^n b_j$ denote the n th partial sums. As $\sum a_n$ and $\sum b_n$ converge, we know that the sequences $\{s_n\}$ and $\{t_n\}$ converge.

Note that the partial sums $\sum_{j=1}^n (a_j + b_j)$ are equal to $s_n + t_n$. By the limit law for addition, we conclude that the sequence $\{s_n + t_n\}_{n \geq 1}$ converges and $\lim(s_n + t_n) = \lim s_n + \lim t_n$. By definition, this is exactly what it means for the series $\sum (a_n + b_n)$ to converge to the number $\sum a_n + \sum b_n$. $\square$

Problem 3. (a) Prove that

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}.$$

(b) Use part (a) to compute

$$\sum_{n=1}^{\infty} \frac{n}{2^n}.$$

Solution: (a) Let $s_n = \sum_{k=1}^n \frac{k-1}{2^{k+1}}$.

We will prove that

$$s_n = \frac{1}{2} - \frac{n+1}{2^{n+1}} \quad \text{for } n \geq 1$$

by induction. For the base case, we note that

$$s_1 = \frac{0}{4} = \frac{1}{2} - \frac{2}{4}.$$

For the inductive step, assume that $s_n = \frac{1}{2} - \frac{n+1}{2^{n+1}}$ for some $n \geq 1$. Then

$$s_{n+1} = s_n + \frac{n}{2^{n+2}} = \frac{1}{2} - \frac{n+1}{2^{n+1}} + \frac{n+1}{2^{n+1}} - \frac{n+2}{2^{n+2}} = \frac{1}{2} - \frac{n+2}{2^{n+2}}.$$

Together, we conclude that our formula for s_n for all $n \geq 1$ by induction.

Next, we claim that $\lim_{n \rightarrow \infty} \frac{n}{2^n} = 0$. Notice that $n \leq \left(\frac{3}{2}\right)^n$ for all $n \geq 1$, and so

$$0 \leq \frac{n}{2^n} \leq \left(\frac{3/2}{2}\right)^n = \left(\frac{3}{4}\right)^n.$$

The right-hand side converges to zero as $n \rightarrow \infty$. Therefore, $\frac{n}{2^n}$ also converges to zero by the squeeze theorem.

Altogether, we conclude

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(\frac{1}{2} - \frac{n+1}{2^{n+1}} \right) = \frac{1}{2}.$$

(In the last equality, we noted that $\frac{n+1}{2^{n+1}}$ is a subsequence of $\frac{n}{2^n}$, and therefore also converges to zero.) By definition of a convergent series, we conclude that

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}.$$

(b) Note that

$$\sum_{k=1}^n \frac{k}{2^k} = 2 \sum_{k=1}^n \frac{k-1}{2^{k+1}} + \sum_{k=1}^n \frac{1}{2^k}.$$

By part (a), $\sum \frac{k-1}{2^{k+1}}$ converges to $\frac{1}{2}$, and thus $2 \sum \frac{k-1}{2^{k+1}}$ converges to 1 by problem 2. From lecture, we know that the geometric series $\sum \frac{1}{2^k}$ converges to $\frac{1}{1-(1/2)} - 1 = 1$. (Notice that we are starting at $k = 1$, not $k = 0$.) Together, we conclude $\sum_{k=1}^{\infty} \frac{k}{2^k}$ converges to $1 + 1 = 2$. $\square$

Problem 5. (a) Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely, then the series $\sum_{n=1}^{\infty} a_n^2$ also converges.

(b) Provide an example where $\sum_{n=1}^{\infty} a_n$ diverges and $\sum_{n=1}^{\infty} a_n^2$ converges.

Solution: (a) Note that $x^2 \leq |x|$ for any $x \in [-1, 1]$. (I recommend that you draw the graphs of x^2 and $|x|$ to check that this is true.)

As $\sum a_n$ converges, then we have $\lim a_n = 0$, and so there exists an N so that $|a_n| < 1$ for all $n \geq N$. By the previous paragraph, this means that $a_n^2 \leq |a_n|$ for all $n \geq N$. As $\sum |a_n|$ converges, then $\sum_{n=N}^{\infty} a_n^2$ also converges by the comparison test. Adding the first $N-1$ terms back in, we conclude $\sum_{n=1}^{\infty} a_n^2$ converges as well.

(b) Consider $a_n = \frac{1}{n}$. In lecture, we proved that $\sum \frac{1}{n}$ diverges and $\sum \frac{1}{n^2}$ converges. $\square$

Problem 6. For each of the following series, determine if it converges and prove your answer.

$$(a) \sum_{n=2}^{\infty} \frac{1}{[n + (-1)^n]^2} \quad (b) \sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n}) \quad (c) \sum_{n=1}^{\infty} (-1)^n$$

Solution: (a) We will prove that this series **converges**. For $n \geq 2$, we have

$$\frac{1}{[n + (-1)^n]^2} \leq \frac{1}{(n-1)^2}.$$

From lecture, we know that the series

$$\sum_{n=2}^{\infty} \frac{1}{(n-1)^2} = \sum_{k=1}^{\infty} \frac{1}{k^2}$$

converges. Therefore our series $\sum \frac{1}{[n+(-1)^n]^2}$ also converges by the comparison test.

(b) We will prove that this series **diverges**. Let $s_n = \sum_{k=1}^n (\sqrt{k+1} - \sqrt{k})$ denote the n th partial sum.

Claim: $s_n = \sqrt{n+1} - 1$ for all $n \geq 1$. We will argue using induction.

- Base case: For $n = 1$, we have

$$s_1 = \sqrt{2} - \sqrt{1} = \sqrt{2} - 1.$$

- Inductive step: Suppose that $s_n = \sqrt{n+1} - 1$ for some $n \geq 1$. Then the formula for s_{n+1} holds, since:

$$\begin{aligned} s_{n+1} &= s_n + (\sqrt{n+2} - \sqrt{n+1}) \\ &= (\sqrt{n+1} - 1) + (\sqrt{n+2} - \sqrt{n+1}) = \sqrt{n+2} - 1. \end{aligned}$$

By induction, we conclude that $s_n = \sqrt{n+1} - 1$ holds for all $n \geq 1$.

Notice that the sequence $s_n = \sqrt{n+1} - 1$ diverges to $+\infty$. Indeed, given $M > 0$, pick $N \in \mathbb{N}$ so that $N > (M+1)^2$, and then for any $n \geq N$ we have

$$\sqrt{n+1} - 1 \geq \sqrt{N} - 1 > \sqrt{(M+1)^2} - 1 = M.$$

As the sequence $\{s_n\}$ diverges, then the series $\sum(\sqrt{n+1} - \sqrt{n})$ diverges.

(c) We will prove that this series **diverges**. Suppose towards a contradiction that $\sum(-1)^n$ converges. Then $\lim(-1)^n = 0$ by Lecture 16. However, we know that the sequence $\{(-1)^n\}$ diverges by Lecture 10, and so this is a contradiction. $\square$

Math 521, Section 2
Homework 7
Solutions

Problem 1. For each of the following series, determine if it converges and prove your answer.

$$(a) \sum_{n=2}^{\infty} \frac{1}{\log n} \qquad (b) \sum_{n=1}^{\infty} \frac{n-1}{n^2} \qquad (c) \sum_{n=1}^{\infty} \frac{n^2}{n!}$$

Solution: (a) We'll use the dyadic criterion. Consider the series $\sum_{k=1}^{\infty} a_k$ with $a_k = \frac{1}{\log(k+1)}$ for $k \geq 1$. (Strictly speaking, in order to apply the dyadic criterion we need a series that starts at $n = 1$, so we reindexed.)

Then $\{a_k\}_{k \geq 1}$ is a decreasing sequence of nonnegative numbers. Therefore, by the dyadic criterion, $\sum_{k=1}^{\infty} a_k$ converges if and only if the series $\sum_{n=0}^{\infty} 2^n a_{2^n}$ converges. Note that

$$2^n a_{2^n} = \frac{2^n}{\log(2^n + 1)} \geq \frac{2^n}{\log(2^{n+1})} = \frac{1}{\log 2} \cdot \frac{2^n}{n+1}.$$

Now, the sequence $\frac{2^n}{n+1}$ diverges to $+\infty$, and so it cannot converge to 0. Multiplying this by the constant $\frac{1}{\log 2}$, we see that $2^n a_{2^n}$ also cannot converge to 0. Therefore the series $\sum 2^n a_{2^n}$ diverges, and thus $\sum a_k$ **diverges** as well.

(b) We'll use the comparison test. For $n \geq 2$, we have

$$n-1 \geq n - \frac{n}{2} = \frac{n}{2} \implies \frac{n-1}{n^2} \geq \frac{n/2}{n^2} = \frac{1}{2n}.$$

From lecture we know that the series $\sum \frac{1}{n}$ diverges, and thus $\sum_{n=2}^{\infty} \frac{1}{2n}$ diverges as well. Therefore, by the comparison test, $\sum_{n=2}^{\infty} \frac{n-1}{n^2}$ also **diverges**. Note that $\frac{n-1}{n^2}$ is equal to zero for $n = 1$, so $\sum_{n=2}^{\infty} \frac{n-1}{n^2}$ and $\sum_{n=1}^{\infty} \frac{n-1}{n^2}$ are the same series.

(c) We'll use the ratio test. Set $a_n = \frac{n^2}{n!}$. Then

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)^2 \cdot n!}{n^2 \cdot (n+1)!} = \frac{n+1}{n^2} \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Therefore

$$\limsup \left| \frac{a_{n+1}}{a_n} \right| = \lim \left| \frac{a_{n+1}}{a_n} \right| = 0 < 1,$$

and so the series **converges** by the ratio test. □

Problem 2. For each of the following series, determine if it converges and prove your answer.

$$(a) \sum_{n=1}^{\infty} \frac{n^4}{4^n} \quad (b) \sum_{n=1}^{\infty} \frac{n!}{n^4 + 3} \quad (c) \sum_{n=1}^{\infty} \frac{2^n}{n!}$$

Solution: (a) We'll use the ratio test. Set $a_n = \frac{n^4}{4^n}$. Then

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)^4 \cdot 4^n}{n^4 \cdot 4^{n+1}} = \frac{(n+1)^4}{4n^4} \longrightarrow \frac{1}{4} \quad \text{as } n \rightarrow \infty.$$

Therefore $\limsup \left| \frac{a_{n+1}}{a_n} \right| = \frac{1}{4} < 1$, and so the series **converges** by the ratio test.

(b) This series **diverges**. In brief, the sequence $\frac{n!}{n^4+3}$ diverges to $+\infty$, and so it cannot converge to zero. More concretely, notice that for $n \geq 5$ we have

$$n! \geq n(n-1)(n-2)(n-3)(n-4),$$

and so

$$\frac{n^4 + 3}{n!} \leq \frac{n^4 + 3}{n(n-1)(n-2)(n-3)(n-4)} \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Therefore the sequence $\frac{n!}{n^4+3} \rightarrow +\infty$ by HW5#1. In particular, there must exist an N so that $\frac{n!}{n^4+3} \geq 1$ for all $n \geq N$, and so $\frac{n!}{n^4+3}$ cannot converge to 0.

(c) We'll use the ratio test. Set $a_n = \frac{2^n}{n!}$. Then

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{2^{n+1} \cdot n!}{(n+1)! \cdot 2^n} = \frac{2}{n+1} \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Therefore $\limsup \left| \frac{a_{n+1}}{a_n} \right| = 0 < 1$, and so the series **converges** by the ratio test. □

Problem 3. For each of the following series, determine if it converges and prove your answer.

$$(a) \sum_{n=1}^{\infty} \frac{1}{n^n} \quad (b) \sum_{n=1}^{\infty} \frac{n!}{n^n} \quad (c) \sum_{n=1}^{\infty} \frac{(-1)^n n!}{2^n}$$

Solution: (a) We'll use the root test. Set $a_n = \frac{1}{n^n}$. Then

$$|a_n|^{\frac{1}{n}} = \frac{1}{n} \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Therefore $\limsup |a_n|^{\frac{1}{n}} = 0 < 1$, and so the series **converges** by the root test.

(b) We'll use the root test. Set $a_n = \frac{n!}{n^n}$, and consider the sequence

$$|a_n|^{\frac{1}{n}} = \left(\frac{n!}{n^n} \right)^{\frac{1}{n}} = \frac{(n!)^{1/n}}{n}.$$

In Lecture 15 we showed that this sequence converges to $1/e$. Therefore

$$\limsup |a_n|^{\frac{1}{n}} = \frac{1}{e} < 1,$$

and so the series **converges** by the root test.

(c) This series **diverges**, because the sequence $\frac{(-1)^n n!}{2^n}$ does not converge to 0. Specifically, we know that the sequence $\frac{2^n}{n!}$ converges to 0. (Indeed, we proved in problem 2c that the series $\sum \frac{2^n}{n!}$ converges, and so its terms must converge to 0). So, by HW5#1, the sequence $\frac{n!}{2^n}$ diverges to $+\infty$. In particular, there exists an N so that $\frac{n!}{2^n} \geq 1$ for all $n \geq N$. Therefore $|\frac{(-1)^n n!}{2^n}| \geq 1$ for all $n \geq N$, and so the terms $\frac{(-1)^n n!}{2^n}$ in our series cannot converge to 0. $\square$

Problem 4. Consider the series

$$\sum_{n=1}^{\infty} 2^{(-1)^n - n} = \frac{1}{4} + \frac{1}{2} + \frac{1}{16} + \frac{1}{8} + \frac{1}{64} + \frac{1}{32} + \dots$$

- (a) Show that the ratio test is inconclusive.
- (b) Prove that the series converges using the root test.

Solution: (a) Set $a_n = 2^{(-1)^n - n}$. Then the sequence $\left| \frac{a_{n+1}}{a_n} \right|$ is

$$\frac{1/2}{1/4} = 2, \quad \frac{1/16}{1/2} = \frac{1}{8}, \quad \frac{1/8}{1/16} = 2, \quad \frac{1/64}{1/8} = \frac{1}{8}, \quad \dots$$

In other words, the even terms of this sequence are $\frac{1}{8}$ and the odd terms are 2. Therefore

$$\liminf \left| \frac{a_{n+1}}{a_n} \right| = \frac{1}{8}, \quad \limsup \left| \frac{a_{n+1}}{a_n} \right| = 2,$$

and so

$$\liminf \left| \frac{a_{n+1}}{a_n} \right| \leq 1 \leq \limsup \left| \frac{a_{n+1}}{a_n} \right|.$$

(b) Set $a_n = 2^{(-1)^n - n}$. Then the sequence $|a_n|^{\frac{1}{n}}$ is

$$\frac{1}{2^2}, \quad \frac{1}{2^{\frac{1}{2}}}, \quad \frac{1}{2^{\frac{4}{3}}}, \quad \frac{1}{2^{\frac{3}{4}}}, \quad \frac{1}{2^{\frac{6}{5}}}, \quad \frac{1}{2^{\frac{5}{6}}}, \quad \dots$$

For n even, we have

$$|a_n|^{\frac{1}{n}} = \frac{1}{(2^{n-1})^{\frac{1}{n}}} = \frac{1}{2^{1-\frac{1}{n}}} = \frac{1}{2} \cdot 2^{\frac{1}{n}} \longrightarrow \frac{1}{2} \quad \text{as } n \rightarrow \infty.$$

(Recall from Lecture 15 that $r^{\frac{1}{n}} \rightarrow 1$ for any $r > 0$.) For n odd, we have

$$|a_n|^{\frac{1}{n}} = \frac{1}{(2^{n+1})^{\frac{1}{n}}} = \frac{1}{2^{1+\frac{1}{n}}} = \frac{1}{2} \cdot \frac{1}{2^{\frac{1}{n}}} \longrightarrow \frac{1}{2} \quad \text{as } n \rightarrow \infty.$$

Therefore

$$\lim_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \frac{1}{2} < 1,$$

and so the series converges by the root test. □

Problem 5. Below are two series $\sum_{n=1}^{\infty} a_n$ and the numbers $s \in \mathbb{R}$ to which they converge. For each of these series, what is the set of all numbers $\tilde{s} \in \mathbb{R}$ such that there exists a rearrangement that satisfies $\sum_{n=1}^{\infty} \tilde{a}_n = \tilde{s}$? Prove your answer.

$$(a) \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{n} = -\log 2 \qquad (b) \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{\pi^2}{12}$$

Solution: (a) We claim that the set is $\mathbb{R}$. Note that:

- $\sum \frac{(-1)^n}{n}$ converges: by the alternating series test, since $\{\frac{1}{n}\}$ is a decreasing sequence that converges to 0
- $\sum \frac{(-1)^n}{n}$ does not converge absolutely: since $|\frac{(-1)^n}{n}| = \frac{1}{n}$, and we know $\sum \frac{1}{n}$ diverges by Lecture 11

Fix $\tilde{s} \in \mathbb{R}$. Applying the Riemann rearrangement result (with $\alpha = \tilde{s} = \beta$), we deduce that there exists a rearrangement $\sum \tilde{a}_n$ that converges to $\tilde{s}$. This works for any $\tilde{s} \in \mathbb{R}$, and so $\mathbb{R}$ is the desired set.

(b) We claim that the set is $\{-\frac{\pi^2}{12}\}$. Note that $\sum \frac{(-1)^n}{n^2}$ converges absolutely, since $|\frac{(-1)^n}{n^2}| = \frac{1}{n^2}$ and we know $\sum \frac{1}{n^2}$ converges by Lecture 16. Suppose that $\sum \tilde{a}_n$ is a rearrangement that converges to some $\tilde{s} \in \mathbb{R}$. Applying the second theorem from Lecture 20, we deduce that $\tilde{s} = -\frac{\pi^2}{12}$. This works for any rearrangement, so $-\frac{\pi^2}{12}$ is the only value that can be attained. $\square$

Problem 6. Prove or disprove the following statements:

- (a) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_{2n}$ converges.
- (b) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_{2n}$ converges absolutely.

Solution: (a) **Disprove:** Consider the series $a_n = \frac{(-1)^n}{n}$. Then:

- $\sum a_n$ converges: by the alternating series test, since $\frac{1}{n}$ is a decreasing sequence that converges to 0.
- $\sum a_{2n} = \sum \frac{1}{2n}$ diverges: This is $\frac{1}{2}$ times the series $\sum \frac{1}{n}$ which we know diverges by lecture.

(b) **Prove:** We will use the Cauchy criterion. Fix $\epsilon > 0$. As $\sum |a_n|$ converges, then there exists an N so that

$$\sum_{k=m}^n |a_k| < \epsilon \quad \text{for all } n \geq m \geq N.$$

Then for $n \geq m \geq N$, we have

$$\sum_{k=m}^n |a_{2k}| \leq \sum_{j=2m}^{2n} |a_j| < \epsilon.$$

(Note that first sum above consists of only even terms $|a_{2m}| + |a_{2m+2}| + \cdots + |a_{2n}|$, which differs from the second sum $|a_{2m}| + |a_{2m+1}| + |a_{2m+2}| + \cdots + |a_{2n}|$ by only the odd terms, each of which is nonnegative.) Therefore $\sum |a_{2n}|$ converges by the Cauchy criterion. $\square$

Math 521, Section 2
Homework 8
Selected Solutions

Problem 1. Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. For each of the following sets $A \subseteq \mathbb{R}$, find its interior and prove your answer.

- (a) $A = [0, 1)$
- (b) $A = \mathbb{Z}$
- (c) $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Solution: (a) Claim: $A^\circ = (0, 1)$. As A° is the largest open subset of A , and $(0, 1)$ is one open subset, we must have $(0, 1) \subseteq A^\circ \subseteq [0, 1)$. Note that 0 is not an interior point: given $r > 0$ we have $(-r, r) \not\subseteq [0, 1)$, since $-\frac{r}{2}$ is a point in $(-r, r)$ that is not in $[0, 1)$.

(b) Claim: $A^\circ = \emptyset$. Fix $n \in \mathbb{Z}$; we will show that n is not an interior point. Given $r > 0$ we have $(n - r, n + r) \not\subseteq \mathbb{Z}$, since $\min\{n + \frac{r}{2}, n + \frac{1}{2}\}$ is in $(n - r, n + r)$ but not in $\mathbb{Z}$.

(c) Claim: $A^\circ = \emptyset$. Fix $a \in A$; we will show that a is not an interior point. As the irrational numbers are dense in $\mathbb{R}$, then for any $r > 0$ we can find a point $y \in \mathbb{Q}^c$ between $a - r$ and a . This y is a point in $(a - r, a + r)$ that is not in A , and so $(a - r, a + r) \not\subseteq A$. $\square$

Problem 2. Recall that d_1 and d_∞ are both metrics on $\mathbb{R}^2$.

- (a) Prove that $d_\infty(x, y) \leq d_1(x, y) \leq 2d_\infty(x, y)$ for all $x, y \in \mathbb{R}^2$.
- (b) Prove that for any $x \in \mathbb{R}^2$ and $r > 0$ we have

$$\{y \in \mathbb{R}^2 : d_\infty(x, y) < \frac{r}{2}\} \subseteq \{y \in \mathbb{R}^2 : d_1(x, y) < r\} \subseteq \{y \in \mathbb{R}^2 : d_\infty(x, y) < r\}.$$

I recommend that you draw a picture of these three sets.

- (c) Prove that a set $A \subseteq \mathbb{R}^2$ is open in the metric space $(\mathbb{R}^2, d_1)$ if and only if it is open in the metric space $(\mathbb{R}^2, d_\infty)$.

Solution: (a) Fix $x = (x_1, x_2)$ and $y = (y_1, y_2)$ in $\mathbb{R}^2$. Then

$$\begin{aligned} |x_1 - y_1| &\leq |x_1 - y_1| + |x_2 - y_2| = d_1(x, y), \\ |x_2 - y_2| &\leq |x_1 - y_1| + |x_2 - y_2| = d_1(x, y), \end{aligned}$$

and so

$$\max\{|x_1 - y_1|, |x_2 - y_2|\} \leq d_1(x, y).$$

This proves the first inequality: $d_\infty(x, y) \leq d_1(x, y)$. On the other hand, note that

$$\begin{aligned} |x_1 - y_1| &\leq \max\{|x_1 - y_1|, |x_2 - y_2|\} = d_\infty(x, y), \\ |x_2 - y_2| &\leq \max\{|x_1 - y_1|, |x_2 - y_2|\} = d_\infty(x, y), \end{aligned}$$

and so

$$|x_1 - y_1| + |x_2 - y_2| \leq 2d_\infty(x, y).$$

This proves that $d_1(x, y) \leq 2d_\infty(x, y)$.

(b) Let's start with the first containment:

$$\{y \in \mathbb{R}^2 : d_\infty(x, y) < \frac{r}{2}\} \subseteq \{y \in \mathbb{R}^2 : d_1(x, y) < r\}.$$

Fix $y \in \mathbb{R}^2$ so that $d_\infty(x, y) < \frac{r}{2}$. Then by the second inequality in part (a) we have

$$d_1(x, y) \leq 2d_\infty(x, y) < 2 \cdot \frac{r}{2} = r,$$

and so $d_1(x, y) < r$.

Next, we turn to the second containment:

$$\{y \in \mathbb{R}^2 : d_1(x, y) < r\} \subseteq \{y \in \mathbb{R}^2 : d_\infty(x, y) < r\}.$$

Fix $y \in \mathbb{R}^2$ so that $d_1(x, y) < r$. Then by the first inequality in part (a), we have

$$d_\infty(x, y) \leq d_1(x, y) < r,$$

and so $d_\infty(x, y) < r$.

(c) $\boxed{\Leftarrow}$ Suppose that A is open in $(\mathbb{R}^2, d_\infty)$. Fix $a \in A$. Then there exists $r > 0$ so that

$$B_r^\infty(a) = \{x \in \mathbb{R}^2 : d_\infty(x, a) < r\} \subseteq A.$$

Note that $B_r^1(a) \subseteq B_r^\infty(a)$ by part (b), and so $B_r^1(a) \subseteq B_r^\infty(a) \subseteq A$. As $a \in A$ was arbitrary, we conclude that A is open in $(\mathbb{R}^2, d_1)$.

$\boxed{\Rightarrow}$ Suppose that A is open in $(\mathbb{R}^2, d_1)$. Fix $a \in A$. Then there exists $r > 0$ so that $B_r^1(a) \subseteq A$. Note that $B_{r/2}^\infty(a) \subseteq B_r^1(a)$ by part (b), and so $B_{r/2}^\infty(a) \subseteq B_r^1(a) \subseteq A$. As $a \in A$ was arbitrary, we conclude that A is open in $(\mathbb{R}^2, d_\infty)$. $\square$

Problem 3. Let X be a nonempty set, and recall that the discrete metric d is a metric on X . Prove that every set $A \subseteq X$ is open in the metric space (X, d) .

Solution: Fix $A \subseteq X$ and $a \in A$. Notice that

$$B_{1/2}(a) = \{x \in X : d(x, a) < \tfrac{1}{2}\} = \{a\}.$$

(Indeed, $d(x, a)$ is either equal to 0 or 1, and so the only time that $d(x, a) < \frac{1}{2}$ is when $d(x, a) = 0$.) Then $a \in B_{1/2}(a) \subseteq A$, and so a is an interior point for A . As $a \in A$ was arbitrary, we conclude that A is open. $\square$

Problem 4. Let (X, d) be any metric space. Define the function $\tilde{d} : X \times X \rightarrow \mathbb{R}$ by

$$\tilde{d}(x, y) = \min\{d(x, y), 1\}.$$

- (a) Prove that $\tilde{d}$ is also a metric on X .
- (b) Prove that a set $A \subseteq X$ is open in (X, d) if and only if it is open in $(X, \tilde{d})$.

Solution: (a) There are 4 properties to check.

1. $\tilde{d}(x, y) \geq 0$ for all $x, y \in X$. Note that $\tilde{d}(x, y)$ is equal to 1 or $d(x, y)$. Both of these are nonnegative; we know $d(x, y) \geq 0$ since d is a metric on X .

2. $\tilde{d}(x, y) = 0 \iff x = y$. If $x = y$, then $d(x, y) = 0$, so $\tilde{d}(x, y) = \min\{0, 1\} = 0$. Conversely, if $\tilde{d}(x, y) = 0$, then $\tilde{d}(x, y) \neq 1$ and so we must have $d(x, y) = \tilde{d}(x, y) = 0$. As d is a metric, this can only mean that $x = y$.

3. $\tilde{d}(x, y) = \tilde{d}(y, x)$ for all $x, y \in X$. As d is a metric, we know that $d(x, y) = d(y, x)$, and so

$$\tilde{d}(x, y) = \min\{d(x, y), 1\} = \min\{d(y, x), 1\} = \tilde{d}(y, x).$$

4. $\tilde{d}(x, z) \leq \tilde{d}(x, y) + \tilde{d}(y, z)$ for all $x, y, z \in X$. We distinguish 4 possible cases:

- $d(x, y) < 1$ and $d(y, z) < 1$: Then

$$\tilde{d}(x, z) \leq d(x, z) \leq d(x, y) + d(y, z) = \tilde{d}(x, y) + \tilde{d}(y, z).$$

- $d(x, y) < 1$ and $d(y, z) \geq 1$: Then

$$\tilde{d}(x, z) \leq 1 \leq d(x, y) + 1 = \tilde{d}(x, y) + \tilde{d}(y, z).$$

- $d(x, y) \geq 1$ and $d(y, z) < 1$: Then

$$\tilde{d}(x, z) \leq 1 \leq 1 + d(y, z) = \tilde{d}(x, y) + \tilde{d}(y, z).$$

- $d(x, y) \geq 1$ and $d(y, z) \geq 1$: Then

$$\tilde{d}(x, z) \leq 1 \leq 1 + 1 = \tilde{d}(x, y) + \tilde{d}(y, z).$$

In all cases, the triangle inequality holds.

- (b) $\boxed{\Leftarrow}$ Suppose that A is open in $(X, \tilde{d})$. Fix $a \in A$. Then there exists $r > 0$ so that

$$B_r^{\tilde{d}}(a) = \{x \in X : \tilde{d}(x, a) < r\} \subseteq A.$$

Note that $B_r^d(a) \subseteq B_r^{\tilde{d}}(a)$, since

$$x \in B_r^d(a) \implies \tilde{d}(x, a) \leq d(x, a) < r \implies x \in B_r^{\tilde{d}}(a).$$

So $B_r^d(a) \subseteq B_r^{\tilde{d}}(a) \subseteq A$. As $a \in A$ was arbitrary, we conclude that A is open in (X, d) .

$\boxed{\Rightarrow}$ Suppose that A is open in (X, d) . Fix $a \in A$. Then there exists $r_1 > 0$ so that $B_{r_1}^d(a) \subseteq A$. Set $r_2 = \min\{r_1, \frac{1}{2}\}$. Note that $B_{r_2}^{\tilde{d}}(a) \subseteq B_{r_1}^d(a)$, since

$$\begin{aligned} x \in B_{r_2}^{\tilde{d}}(a) &\implies \tilde{d}(x, a) = \min\{d(x, a), 1\} < r_2 \leq \frac{1}{2} \\ &\implies \tilde{d}(x, a) = d(x, a) \implies d(x, a) < r_2 \leq r_1. \end{aligned}$$

So $B_{r_2}^{\tilde{d}}(a) \subseteq B_{r_1}^d(a) \subseteq A$. As $a \in A$ was arbitrary, we conclude that A is open in $(X, \tilde{d})$. $\square$

Problem 5. Let (X, d) be a metric space, $A \subseteq X$, and $x \in X$. Prove that x is in A° if and only if there exists an open set U so that $x \in U \subseteq A$.

Solution: $\Rightarrow$ Suppose that $x \in A^\circ$. Then there exists $r > 0$ such that $B_r(x) \subseteq A$. Notice that $U = B_r(x)$ is an open set that satisfies $x \in U \subseteq A$.

$\Leftarrow$ Suppose that $x \in U \subseteq A$ for some open set U . As U is open, there exists $r > 0$ so that $B_r(x) \subseteq U$. Together we have $B_r(x) \subseteq U \subseteq A$, and so $x \in A^\circ$. $\square$

Math 521, Section 2
Homework 9
Solutions

Problem 1. Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. For each of the following sets $A \subseteq \mathbb{R}$, find its closure, isolated points, and limit points. Prove your answer.

- (a) $A = [0, 1)$
- (b) $A = \mathbb{Z}$
- (c) $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Solution: (a) Closure: $\overline{A} = [0, 1]$. As $\overline{A}$ is the smallest closed set containing A , and $[0, 1]$ is one such closed set, we must have $[0, 1) \subseteq \overline{A} \subseteq [0, 1]$. Note that 1 must be in the closure of A : given $r > 0$ we have $(1 - r, 1 + r) \cap [0, 1) \neq \emptyset$, since $\max\{1 - \frac{r}{2}, 0\}$ is in $(1 - r, 1 + r)$ and $[0, 1)$.

Isolated points: $\emptyset$. Fix $a \in A$. Given $r > 0$ we have $(a - r, a + r) \cap A \supsetneq \{a\}$, since $\min\{a + \frac{r}{2}, \frac{a+1}{2}\}$ is a point in $(a - r, a + r) \cap A$ other than a .

Limit points: $A' = [0, 1]$. By definition, $\overline{A}$ is the union of isolated points and limit points, and we just showed that there are no isolated points.

(b) Closure: $\overline{A} = \mathbb{Z}$. We automatically have $\overline{A} \supseteq \mathbb{Z}$. Fix $x \in \mathbb{Z}^c$; we will show that x is not an adherent point. Let $n \in \mathbb{Z}$ so that $n < x < n + 1$. Then $r = \min\{x - n, n + 1 - x\}$ is a positive number so that $(x - r, x + r) \cap \mathbb{Z} = \emptyset$.

Isolated points: $\mathbb{Z}$. Fix $n \in \mathbb{Z}$. Then $(n - 1, n + 1) \cap \mathbb{Z} = \{n\}$.

Limit points: $A' = \emptyset$. By definition, $\overline{A}$ is the union of isolated points and limit points, and we just showed that every point is an isolated point.

(c) Closure: $\overline{A} = [0, 1]$. Note that 0 is an adherent point for A : given $r > 0$, pick $n > \max\{\frac{1}{r}, 1\}$ and then $\frac{1}{n}$ is a point in $(-r, r) \cap A$. Now fix $x \in (0, 1]$; we will show that x is an adherent point. As the rational numbers are dense in $\mathbb{R}$, given $r > 0$ we can find a point $q \in \mathbb{Q}$ between $\max\{x - r, 0\}$ and x , and this point q will be in $(x - r, x + r) \cap A$.

Isolated points: $\emptyset$. Given $x \in A$ and $r > 0$, we pick $q \in \mathbb{Q}$ as in the previous paragraph. This q is a point in $(x - r, x + r) \cap A$ other than x .

Limit points: $A' = [0, 1]$. By definition, $\overline{A}$ is the union of isolated points and limit points, and we just showed that there are no isolated points. □

Problem 2. Let (X, d) be a metric space, let A be a nonempty subset of X , and let U be an open subset of X .

- (a) Prove that $U \cap \overline{A} \subseteq \overline{U \cap A}$.
- (b) Prove that $\overline{U \cap A} = \overline{U} \cap \overline{A}$.
- (c) Conclude that if $U \cap A = \emptyset$, then $U \cap \overline{A} = \emptyset$.

Solution: (a) Fix $x \in U \cap \overline{A}$. We want to show that $x \in \overline{U \cap A}$, i.e., for any $r > 0$, $B_r(x) \cap U \cap A$ is nonempty.

Fix $r > 0$. As $x \in U$ and U is open, there exists $s > 0$ so that $B_s(x) \subseteq U$. Set $\tilde{r} = \min\{r, s\}$. As $x \in \overline{A}$ and $\tilde{r}$ is a positive number, then we know that $B_{\tilde{r}}(x) \cap A$ is nonempty, and so there exists some point $y \in B_{\tilde{r}}(x) \cap A$.

Altogether, we have:

- $y \in A$
- $y \in B_r(x)$: since $y \in B_{\tilde{r}}(x)$, and so $d(y, x) < \tilde{r} \leq r$ and thus $y \in B_r(x)$.
- $y \in U$: since $y \in B_{\tilde{r}}(x)$, and so $d(y, x) < \tilde{r} \leq s$ and thus $y \in B_s(x) \subseteq U$.

So y is a point in $B_r(x) \cap U \cap A$, as desired.

- (b) Recall that $B \subseteq C \implies \overline{B} \subseteq \overline{C}$. Applying this to part (a), we see that

$$\overline{U \cap \overline{A}} \subseteq \overline{U \cap A}.$$

On the other hand, we have $A \subseteq \overline{A}$ and so $U \cap A \subseteq U \cap \overline{A}$. Therefore

$$\overline{U \cap A} \subseteq \overline{U \cap \overline{A}}.$$

Together, this shows that $\overline{U \cap \overline{A}} = \overline{U \cap A}$.

- (c) For any set B , we have $B \subseteq \overline{B}$. So, if $\overline{B} = \emptyset$, then we must have $B = \emptyset$. Therefore:

$$\begin{aligned} U \cap A = \emptyset &\implies \overline{U \cap A} = \emptyset \\ &\implies \overline{U \cap \overline{A}} = \emptyset && \text{by (b)} \\ &\implies U \cap \overline{A} = \emptyset. \end{aligned}$$

□

Problem 3. Recall that d_2 is a metric on $\mathbb{R}^2$. Consider the subspace

$$Y = \{(x, 0) : x \in \mathbb{R}\}$$

with the induced metric d_Y .

- (a) Provide an example of a set $A \subseteq Y$ where A is open in (Y, d_Y) but not open in $(\mathbb{R}^2, d_2)$.
- (b) Prove that for any $B \subseteq Y$, if B is closed in (Y, d_Y) then B is closed in $(\mathbb{R}^2, d_2)$.

Solution: (a) Consider the set $A = \{(x_1, 0) : -1 < x_1 < 1\}$.

First, we will show that A is open in (Y, d_Y) . Fix $x = (x_1, 0) \in A$. Set $r = 1 - |x_1|$. Then $r > 0$ and

$$B_r^Y(x) = \{y \in Y : d_Y(x, y) < r\} \subseteq A,$$

since

$$\begin{aligned} y \in B_r^Y(x) &\implies r > d_Y(x, y) = \sqrt{|x_1 - y_1|^2 + |0 - 0|^2} = |x_1 - y_1| \\ &\implies |y_1| \leq |x_1| + |y_1 - x_1| < |x_1| + r = 1 \\ &\implies y \in A. \end{aligned}$$

Next, we will show that A is not open in $(\mathbb{R}^2, d_2)$ by demonstrating that $(0, 0) \in A$ is not an interior point. Fix $r > 0$. Then

$$B_r^{\mathbb{R}^2}((0, 0)) = \{y \in \mathbb{R}^2 : d_2(y, (0, 0)) < r\} \not\subseteq A,$$

since $(0, \frac{r}{2})$ is a point that is in $B_r^{\mathbb{R}^2}((0, 0))$ but not in A .

(b) By the corollary from Lecture 24, it suffices to show that Y is closed in $(\mathbb{R}^2, d_2)$. Fix $x = (x_1, x_2) \in Y^c$. Then $x_2 \neq 0$. Set $r = |x_2|$. Then $r > 0$ and

$$B_r^{\mathbb{R}^2}(x) = \{y \in \mathbb{R}^2 : d_2(x, y) < r\} \subseteq Y^c,$$

since

$$\begin{aligned} y \in B_r^{\mathbb{R}^2}(x) &\implies r > d_2(x, y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2} \geq |x_2 - y_2| \\ &\implies |y_2| \geq |x_2| - |x_2 - y_2| > |x_2| - r = 0 \\ &\implies y_2 \neq 0 \implies y \in Y^c. \end{aligned}$$

As $x \in Y^c$ was arbitrary, we conclude that Y^c is open, and so Y is closed. □

Problem 4. Let (X, d) be a metric space, $\{x_n\}_{n \geq 1}$ a sequence in X , and $x \in X$. Prove that the sequence $\{x_n\}_{n \geq 1}$ converges to x if and only if for every open set U that contains x , we have $x_n \in U$ for all but finitely many $n \in \mathbb{N}$.

Solution: $\Rightarrow$ Suppose that $x_n \rightarrow x$ in (X, d) as $n \rightarrow \infty$. Fix an open set $U \subseteq X$ such that $x \in U$. As U is open, then there exists $r > 0$ so that $B_r(x) \subseteq U$. Then, by definition of convergence, we know that there exists an N so that $d(x_n, x) < r$ for all $n \geq N$. Together, we conclude that $x_n \in B_r(x) \subseteq U$ for all $n \geq N$.

$\Leftarrow$ Suppose that for any open set containing x , we have $x_n \in U$ for all but finitely many $n \in \mathbb{N}$. Fix $r > 0$. Applying this fact to the open set $B_r(x)$, we see that $x_n \in B_r(x)$ for all but finitely many $n \in \mathbb{N}$. Let N be larger than the largest index for which $x_n \notin B_r(x)$. Then, for any $n \geq N$, we have $x_n \in B_r(x)$ and thus $d(x_n, x) < r$. As $r > 0$ was arbitrary, we conclude that $\{x_n\}_{n \geq 1}$ converges to x in (X, d) . $\square$

Problem 5. Recall that d_2 is a metric on $\mathbb{R}^2$. Suppose that $x^n = (x_1^n, x_2^n)$ is a sequence of points in $\mathbb{R}^2$.

- (a) Prove that $|x_1^n - x_1| \leq d_2(x^n, x)$ for any $x = (x_1, x_2) \in \mathbb{R}^2$.
- (b) Prove that the sequence $\{x^n\}_{n \geq 1}$ converges in $(\mathbb{R}^2, d_2)$ if and only if the sequences $\{x_1^n\}_{n \geq 1}$ and $\{x_2^n\}_{n \geq 1}$ of real numbers both converge.

Solution: (a) Recall that

$$d_2(x^n, x) = (|x_1^n - x_1|^2 + |x_2^n - x_2|^2)^{\frac{1}{2}}.$$

As $|x_2^n - x_2|^2 \geq 0$, then

$$d_2(x^n, x) \geq (|x_1^n - x_1|^2 + 0)^{\frac{1}{2}} \geq |x_1^n - x_1|.$$

- (b) $\implies$ Suppose that $\{x^n\}_{n \geq 1}$ converges to x in $(\mathbb{R}^2, d_2)$. Then

$$d_2(x^n, x) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

By part (a), we know that

$$0 \leq |x_1^n - x_1| \leq d_2(x^n, x).$$

So, by the squeeze theorem we have

$$|x_1^n - x_1| \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

and so $\{x_1^n\}_{n \geq 1}$ converges to x_1 in $(\mathbb{R}, |\cdot|)$. Similarly, we have

$$0 \leq |x_2^n - x_2| \leq d_2(x^n, x),$$

and so $\{x_2^n\}_{n \geq 1}$ converges to x_2 in $(\mathbb{R}, |\cdot|)$.

- $\impliedby$ Suppose that $\{x_1^n\}_{n \geq 1}$ converges to x_1 and $\{x_2^n\}_{n \geq 1}$ converges to x_2 . Then

$$|x_1^n - x_1| \rightarrow 0 \quad \text{and} \quad |x_2^n - x_2| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

So, by the limit laws we have

$$d_2(x^n, x) = (|x_1^n - x_1|^2 + |x_2^n - x_2|^2)^{\frac{1}{2}} \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

and thus $\{x^n\}_{n \geq 1}$ converges to x in $(\mathbb{R}^2, d_2)$. □

Problem 6. Let (X, d) be a metric space and $\{x_n\}_{n \geq 1}$ a sequence in X . Prove that if $\{x_n\}_{n \geq 1}$ converges to $x \in X$, then for any $y \in X$, the sequence $\{d(x_n, y)\}_{n \geq 1}$ of real numbers converges to $d(x, y)$.

Solution: Fix $y \in X$. By triangle inequality,

$$\begin{aligned} d(x_n, y) &\leq d(x_n, x) + d(x, y) &\implies d(x_n, y) - d(x, y) &\leq d(x_n, x) \\ d(x, y) &\leq d(x, x_n) + d(x_n, y) &\implies d(x, y) - d(x_n, y) &\leq d(x_n, x) \end{aligned}$$

Together, we obtain

$$|d(x_n, y) - d(x, y)| \leq d(x_n, x).$$

As $\{x_n\}_{n \geq 1}$ converges to x in (X, d) , then the right-hand side above converges to 0 as $n \rightarrow \infty$. Therefore, by the squeeze theorem, we conclude that

$$|d(x_n, y) - d(x, y)| \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

and thus $d(x_n, y)$ converges to $d(x, y)$. □

Math 521, Section 2
Homework 10
Selected Solutions

Problem 3. Let (X, d) be a metric space.

- (a) Prove that if $A \subseteq B \subseteq X$, A is closed in X , and B is compact, then A is compact.
- (b) Prove that if $C, D \subseteq X$ are both compact, then $C \cap D$ is compact.

Solution: (a) Let $\{U_i\}_{i \in I}$ be an open cover of A . As A is closed, then A^c is open. So $\{A^c\} \cup \{U_i\}_{i \in I}$ is an open cover for B , since each set is open and we have

$$B \subseteq X = A^c \cup A \subseteq A^c \cup \bigcup_{i \in I} U_i.$$

As B is compact, there must exist a finite subcover for B . We may write this subcover as $\{A^c\} \cup \{U_i\}_{i \in J}$ for some finite set $J \subseteq I$, because if A^c is not part of the subcover, then we can add A^c and still have a finite subcover. Altogether, this yields

$$B \subseteq A^c \cup \bigcup_{i \in J} U_i.$$

As A is a subset of B and is disjoint from A^c , then we must have

$$A \subseteq \bigcup_{i \in J} U_i.$$

So $\{U_i\}_{i \in J}$ is a finite subcover of A .

(b) As C and D are compact, then they are closed by Lecture 27. Therefore $C \cap D$ is closed as well. Altogether, we see that $C \cap D$ is a closed subset of a compact set C , and thus $C \cap D$ is compact by part (a). $\square$

Problem 4. Let (X, d) be a metric space and let $Y \subseteq X$. We call the set Y complete if the subspace Y with the induced metric is complete. Prove that if Y is compact, then Y is complete.

Solution: Let $\{y_n\}_{n \geq 1}$ be a Cauchy sequence in Y . We want to show that $\{y_n\}_{n \geq 1}$ converges to some point in Y .

As Y is compact, then by Lecture 26 there exists a subsequence $\{y_{k_n}\}_{n \geq 1}$ which converges to some point $y \in Y$.

To finish, we will show that the whole sequence $\{y_n\}_{n \geq 1}$ converges to y . Fix $r > 0$. As y_{k_n} converges to y , there exists N_1 so that

$$d(y_{k_n}, y) < \frac{r}{2} \quad \text{for all } n \geq N_1.$$

As $\{y_n\}_{n \geq 1}$ is a Cauchy sequence, then there exists N_2 so that

$$d(y_n, y_m) < \frac{r}{2} \quad \text{for all } n, m \geq N_2.$$

Set $N = \max\{N_1, N_2\}$. Then, for $n \geq N$ we have $k_n \geq n \geq N$, and so

$$d(y_n, y) \leq d(y_n, y_{k_n}) + d(y_{k_n}, y) < \frac{r}{2} + \frac{r}{2} = r.$$

As $r > 0$ was arbitrary, we conclude that $\{y_n\}_{n \geq 1}$ converges to y . □

Math 521, Section 2
Homework 11
Selected Solutions

Problem 2. Let (X, d) be a metric space.

- (a) Prove that if $F \subseteq X$ is disconnected and closed in X , then there exist $A, B \subseteq X$ that are nonempty, separated, and closed in X so that $F = A \cup B$.
- (b) Prove that if $U \subseteq X$ is disconnected and open in X , then there exist $A, B \subseteq X$ that are nonempty, separated, and open in X so that $U = A \cup B$.

Solution: (a) As F is disconnected, then by Lecture 29 we know that there exists a set $A \neq \emptyset, F$ that is both open and closed in F . Set $B = F \setminus A$. We will show that these sets A and B satisfy the desired properties.

- A and B are nonempty: since $A \neq \emptyset$ and $A \neq F$.
- A is closed in X : We know that A is closed in F . As F is closed in X , then A is also closed in X (by Lecture 24).
- B is closed in X : We know that A is open in F , and so $B = F \setminus A$ is closed in F . As F is closed in X , then B must also be closed in X .
- A and B are separated: We know that $A \cap B = \emptyset$, since $B = F \setminus A$. As A is closed in X , then $\overline{A} = A$ and so $\overline{A} \cap B = A \cap B = \emptyset$. Similarly, as B is closed in X , we have $A \cap \overline{B} = A \cap B = \emptyset$.

(b) As U is disconnected, then by Lecture 29 we know that there exists a set $A \neq \emptyset, U$ that is both open and closed in U . Set $B = U \setminus A$. We will show that these sets A and B satisfy the desired properties.

- A and B are nonempty: since $A \neq \emptyset$ and $A \neq U$.
- A is open in X : We know that A is open in U . As U is open in X , then A is also open in X (by Lecture 24).
- B is open in X : We know that A is closed in U , and so $B = U \setminus A$ is open in U . As U is open in X , then B must also be open in X .
- A and B are separated: We know that $A \cap B = \emptyset$, since $B = U \setminus A$. Then

$$\begin{aligned} A \cap B = \emptyset &\implies A \subseteq B^c \\ &\implies \overline{A} \subseteq \overline{B^c} = B^c \\ &\implies \overline{A} \cap B = \emptyset. \end{aligned}$$

(In the second line, we used the fact that B^c is closed in X , since B is open.) Similarly,

$$\begin{aligned} A \cap B = \emptyset &\implies B \subseteq A^c \\ &\implies \overline{B} \subseteq \overline{A^c} = A^c \\ &\implies A \cap \overline{B} = \emptyset. \end{aligned}$$

□

Problem 3. Suppose that (X, d) is a metric space such that for any pair of points $a, b \in X$, there exists a connected set $C \subseteq X$ so that $a, b \in C$. Prove that X is connected.

Solution: Suppose towards a contradiction that X is disconnected. Then there exist sets $A, B \subseteq X$ so that:

- $X = A \cup B$,
- A and B are nonempty,
- A and B are separated.

As A and B are nonempty, then there exists a point $a \in A$ and $b \in B$. The problem statement then tells us that there exists a connected set $C \subseteq X$ so that $a, b \in C$. As C is connected and C is contained in the union of the separated sets A and B , then by Lecture 29 we know that either $C \subseteq A$ or $C \subseteq B$. As $a \in A \cap C$, then we must be in the first case: $C \subseteq A$ and $B \cap C = \emptyset$. But this contradicts that $b \in B \cap C$. $\square$

Problem 4. Let (X, d) be a metric space and $A \subseteq X$. Prove that if A is connected, then $\overline{A}$ is connected.

Solution: Suppose towards a contradiction that $\overline{A}$ is disconnected. Then there exist sets $B, C \subseteq X$ so that:

- $\overline{A} = B \cup C$,
- B and C are nonempty,
- B and C are separated.

As A is connected, then by Lecture 29 either $A \subseteq B$ or $A \subseteq C$. We may assume that $A \subseteq B$ after swapping B and C if necessary. Then $\overline{A} \subseteq \overline{B}$. On the other hand, we know B and C are separated, and so $\overline{B} \cap C = \emptyset$. Together, we conclude that $\overline{A} \cap C = \emptyset$. But $\overline{A} = B \cup C$, and so the only way this can happen is if $C = \emptyset$. This contradicts that C is nonempty. $\square$

Problem 5. Let (X, d) be a metric space and $A, B \subseteq X$. Prove that if A and B are both connected and $A \cap B \neq \emptyset$, then $A \cup B$ is connected.

Solution: Suppose towards a contradiction that $A \cup B$ were disconnected. Then there exist sets $C, D \subseteq X$ so that:

- $A \cup B = C \cup D$,
- C and D are nonempty,
- C and D are separated.

As $A \subseteq C \cup D$ and A is connected, then either $A \subseteq C$ or $A \subseteq D$ by Lecture 29. We may assume that $A \subseteq C$ after swapping C and D if necessary.

Similarly, as $B \subseteq C \cup D$ and B is connected, then either $B \subseteq C$ or $B \subseteq D$. On the other hand, we know that $A \subseteq C$ and $A \cap B \neq \emptyset$, and so at least one point of B is in C . Together, we conclude that all of B is contained in C .

Altogether, we conclude $A \cup B = C$. This contradicts that D is nonempty. $\square$

Problem 6. Let (X, d) be a metric space.

- (a) Let $A, B, C \subseteq X$. Prove that if the sets A and B are separated and the sets A and C are separated, then the sets A and $B \cup C$ are separated.
- (b) Prove that if $F, G \subseteq X$ are closed and $F \cup G$ and $F \cap G$ are connected, then F is connected.

Solution: (a) As A and B are separated, then

$$\overline{A} \cap B = \emptyset \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

Similarly, as A and C are separated, we know

$$\overline{A} \cap C = \emptyset \quad \text{and} \quad A \cap \overline{C} = \emptyset.$$

Together, we see that

$$\overline{A} \cap (B \cup C) = (\overline{A} \cap B) \cup (\overline{A} \cap C) = \emptyset \cup \emptyset = \emptyset,$$

and

$$A \cap \overline{(B \cup C)} = A \cap (\overline{B} \cup \overline{C}) = (A \cap \overline{B}) \cup (A \cap \overline{C}) = \emptyset \cup \emptyset = \emptyset.$$

(Recall that $\overline{B \cup C} = \overline{B} \cup \overline{C}$ by Lecture 23.) Therefore A and $B \cup C$ are separated.

(b) Suppose towards a contradiction that F is disconnected. Then there exist sets $A, B \subseteq X$ so that:

- $F = A \cup B$,
- A and B are nonempty,
- A and B are separated.

As $F \cap G$ is connected and $F \cap G$ is contained in the union of the separated sets A and B , then by Lecture 29 we know that either $F \cap G \subseteq A$ or $F \cap G \subseteq B$. We may assume that $F \cap G \subseteq A$ after swapping A and B if necessary.

Claim: B and G are separated. First, we note that $\overline{B} \cap G = \emptyset$, since

$$\begin{aligned} x \in \overline{B} &\implies x \notin A \quad \text{and} \quad x \in F \\ &\implies x \notin F \cap G \quad \text{and} \quad x \notin G \setminus F \\ &\implies x \notin G. \end{aligned}$$

Next, we use that G is closed to show $B \cap \overline{G} = \emptyset$:

$$\begin{aligned} x \in \overline{G} = G &\implies x \in G \setminus F \quad \text{or} \quad x \in F \cap G \subseteq A \\ &\implies x \notin B. \end{aligned}$$

Together, we conclude B and G are separated.

Therefore, applying part (a) to the sets B, A and B, G , we see that B and $A \cup G$ are separated. Note that B and $A \cup G$ are both nonempty since A and B are, and

$$B \cup A \cup G = F \cup G.$$

Altogether, we conclude that $F \cup G$ is disconnected, which is a contradiction. □

Math 521, Section 2
Homework 12
Selected Solutions

Problem 1. Let (X, d_X) and (Y, d_Y) be metric spaces, $f : X \rightarrow Y$, and $x_0 \in X$. Prove that f is continuous at x_0 if and only if for any open set $V \subseteq Y$ containing $f(x_0)$, there is an open set $U \subseteq X$ containing x_0 so that $x \in U$ implies $f(x) \in V$.

Solution: $\Rightarrow$ Suppose that f is continuous at x_0 . Fix an open set $V \subseteq Y$ so that $f(x_0) \in V$. Then there exists $\epsilon > 0$ so that $B_\epsilon^Y(f(x_0)) \subseteq V$. As f is continuous at x_0 , then there exists $\delta > 0$ so that

$$d_X(x, x_0) < \delta \quad \Longrightarrow \quad d_Y(f(x), f(x_0)) < \epsilon.$$

Set $U = B_\delta^X(x_0)$. Then U is an open subset of X that contains x_0 . Moreover, $x \in U$ implies $f(x) \in V$, since

$$\begin{aligned} x \in U = B_\delta^X(x_0) &\Longrightarrow d_X(x, x_0) < \delta \\ &\Longrightarrow d_Y(f(x), f(x_0)) < \epsilon \\ &\Longrightarrow f(x) \in B_\epsilon^Y(f(x_0)) \subseteq V. \end{aligned}$$

$\Leftarrow$ Assume that for any open set $V \subseteq Y$ containing $f(x_0)$, there is an open set $U \subseteq X$ containing x_0 so that $x \in U$ implies $f(x) \in V$. We want to show that f is continuous at x_0 . Fix $\epsilon > 0$. Then $V = B_\epsilon^Y(f(x_0))$ is an open subset of Y containing $f(x_0)$. So, we know that there exists an open set $U \subseteq X$ containing x_0 so that $x \in U$ implies $f(x) \in V$. As U is open, there must exist $\delta > 0$ so that $B_\delta^X(x_0) \subseteq U$. Altogether, we conclude

$$\begin{aligned} d_X(x, x_0) < \delta &\Longrightarrow x \in B_\delta^X(x_0) \subseteq U \\ &\Longrightarrow f(x) \in V = B_\epsilon^Y(f(x_0)) \\ &\Longrightarrow d_Y(f(x), f(x_0)) < \epsilon \end{aligned}$$

as desired. □

Problem 2. Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \rightarrow Y$ a function. Prove that f is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$ for any $A \subseteq X$.

Solution: $\boxed{\implies}$ Suppose that f is continuous. Fix $A \subseteq X$ and $y \in f(\overline{A})$; we need to show that $y \in \overline{f(A)}$ as well. As $y \in f(\overline{A})$, then there exists $x_0 \in \overline{A}$ so that $f(x_0) = y$. Then, there exists a sequence $\{x_n\}$ in A so that $x_n \rightarrow x_0$ (by Lecture 25). As f is continuous at x_0 , then $f(x_n) \rightarrow f(x_0)$ (by Lecture 30). So $\{f(x_n)\}$ is a sequence in $f(A)$ that converges to $f(x_0)$, and so $f(x_0) \in \overline{f(A)}$.

$\boxed{\impliedby}$ Assume that $f(\overline{A}) \subseteq \overline{f(A)}$ for any $A \subseteq X$. Suppose towards a contradiction that f were not continuous. Then there would exist a sequence $\{x_n\}$ in X so that x_n converges to x_0 but $f(x_n)$ does not converges to $f(x_0)$. Negating the definition of ‘ $f(x_n)$ converges to $f(x_0)$ ’, we get: there exists $\epsilon > 0$ so that for any N , there is some $n \geq N$ so that $d_Y(f(x_n), f(x_0)) \geq \epsilon$.

- Taking $N = 1$, there must exist some $k_1 \geq 1$ so that $d_Y(f(x_{k_1}), f(x_0)) \geq \epsilon$.
- Taking $N = k_1 + 1$, there must exist some $k_2 > k_1$ so that $d_Y(f(x_{k_2}), f(x_0)) \geq \epsilon$.

Proceeding inductively, we get a subsequence $\{x_{k_n}\}$ so that $d_Y(f(x_{k_n}), f(x_0)) \geq \epsilon$ for all $n \geq 1$.

Consider the set $A = \{x_{k_n} : n \in \mathbb{N}\}$, and note that $x_0 \in \overline{A}$ since $x_{k_n} \rightarrow x_0$. For this set we must have $f(\overline{A}) \subseteq \overline{f(A)}$, and so $f(x_0) \in \overline{f(A)} \subseteq \overline{f(A)}$. This implies that $B_\epsilon(f(x_0)) \cap f(A) \neq \emptyset$, and so there exists a point $f(x_{k_n}) \in f(A)$ so that $d_Y(f(x_{k_n}), f(x_0)) < \epsilon$. However, this contradicts how x_{k_n} was chosen. $\square$

Problem 4. Consider the metric spaces $(\mathbb{R}, |\cdot|)$ and $(\mathbb{R}^2, d_2)$.

- (a) Prove that the functions $f, g : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $f(x, y) = x$ and $g(x, y) = y$ are continuous. (Hint: Use sequences.)
- (b) Prove that if $h : [0, 1] \rightarrow \mathbb{R}^2$ is a continuous function such that $h(0) = (x_0, y_0)$, $h(1) = (x_1, y_1)$, and $x_0 < 0 < x_1$, then there exists a point $t \in (0, 1)$ so that $h(t)$ lies on the y -axis.

Solution: (a) Fix a sequence (x_n, y_n) in $(\mathbb{R}^2, d_2)$ that converges to some point (x_0, y_0) . By Lecture 30, it suffices to show that $f(x_n, y_n)$ converges to $f(x_0, y_0)$ and $g(x_n, y_n)$ converges to $g(x_0, y_0)$ in $(\mathbb{R}, |\cdot|)$. Notice that $f(x_n, y_n) = x_n$ and $g(x_n, y_n) = y_n$. As $(x_n, y_n) \rightarrow (x_0, y_0)$ in $(\mathbb{R}^2, d_2)$, then by HW9#5 we know that both $x_n \rightarrow x_0$ and $y_n \rightarrow y_0$ in $(\mathbb{R}, |\cdot|)$.

(b) Write $h(t) = (x(t), y(t)) \in \mathbb{R}^2$. It suffices to show that there exists a point $t \in (0, 1)$ so that $x(t) = 0$. We will argue using the Intermediate Value Theorem. Note that $x : [0, 1] \rightarrow \mathbb{R}$ satisfies $x(0) = x_0 < 0 < x_1 = x(1)$. So it only remains to show that $x(t)$ is continuous.

Claim: $x : [0, 1] \rightarrow \mathbb{R}$ is a continuous function. Note that $x = f \circ h$, where f is the continuous function from part (a). So x is the composition of two continuous functions, and thus is also continuous. $\square$

Problem 5. Prove that the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$f(x, y) = x^2 + y^2 + \sin(x - y)$$

has a global minimum.

Solution: We know that the set $[-1, 1]^2$ is compact in $\mathbb{R}^2$ by Lecture 28. Moreover, the function f is continuous on $[-1, 1]^2$ since it is the sum and composition of polynomials and $\sin$. So, by the extreme value theorem, there must exist a point $(x_0, y_0) \in [-1, 1]^2$ so that

$$f(x, y) \geq f(x_0, y_0) \quad \text{for all } (x, y) \in [-1, 1]^2.$$

For $|x| > 1$, note that

$$f(x, y) = x^2 + y^2 + \sin(x - y) \geq 1 + 0 - 1 = 0.$$

Similarly, when $|y| > 1$ we have

$$f(x, y) = x^2 + y^2 + \sin(x - y) \geq 0 + 1 - 1 = 0.$$

Together, this yields

$$f(x, y) \geq 0 \quad \text{for all } (x, y) \in \mathbb{R}^2 \setminus [-1, 1]^2.$$

On the other hand, notice that $f(0, 0) = 0$. As $(0, 0)$ is a point in $[-1, 1]^2$, then the minimum of f on $[-1, 1]^2$ must be smaller than this:

$$0 = f(0, 0) \geq f(x_0, y_0).$$

Combining this with the previous inequality, we see that

$$f(x, y) \geq f(x_0, y_0) \quad \text{for all } (x, y) \in \mathbb{R}^2 \setminus [-1, 1]^2.$$

Collecting the previous two paragraphs, we conclude that

$$f(x, y) \geq f(x_0, y_0) \quad \text{for all } (x, y) \in \mathbb{R}^2.$$

Therefore (x_0, y_0) is the global minimum of f . □

Problem 6. Let X and Y be metric spaces and $f : X \rightarrow Y$ a function. Prove that if X is compact and f is continuous and bijective, then the inverse function $f^{-1} : Y \rightarrow X$ is also continuous.

Solution: It suffices to prove that for any closed set $F \subseteq X$, the set $(f^{-1})^{-1}(F)$ is closed in Y . Fix $F \subseteq X$ closed. As F is a closed subset of the compact space X , then F is compact as well (by HW10#3). Then $f(F)$ is also compact (by Lecture 32), since f is continuous. In particular, $f(F)$ is a closed subset of Y (by Lecture 27).

Finally, note that the set $f(F)$ is the same as the set $(f^{-1})^{-1}(F)$. Indeed, in the special case where g is a bijective function, then the preimage $g^{-1}(F)$ of F under g is the same thing as the image of F under the inverse function g^{-1} . Here, g is the bijective function f^{-1} , and the inverse function g^{-1} is simply f , since $(f^{-1} \circ f)(x) = x$ for all $x \in X$ and $(f \circ f^{-1})(y) = y$ for all $y \in Y$. $\square$

Math 521, Section 2
Homework 13
Solutions

Problem 1. Consider the sequence of functions $f_n : [0, 1] \rightarrow \mathbb{R}$ given by

$$f_n(x) = \begin{cases} nx & \text{if } x \in [0, \frac{1}{n}], \\ 2 - nx & \text{if } x \in (\frac{1}{n}, \frac{2}{n}], \\ 0 & \text{if } x \in (\frac{2}{n}, 1]. \end{cases}$$

- (a) Prove that f_n converges pointwise to the function $f : [0, 1] \rightarrow \mathbb{R}$, $f(x) = 0$.
- (b) Prove that f_n does not converge uniformly to any function $g : [0, 1] \rightarrow \mathbb{R}$.

Solution: (a) Fix $x \in [0, 1]$. We want to show that the sequence of numbers $f_n(x)$ converges to $f(x) = 0$.

- Case: $x = 0$. Then $f_n(0) = 0$ for all n , and so $f_n(0) \rightarrow 0$.
- Case: $x \in (0, 1]$. Then there exists an $N \in \mathbb{N}$ so that $\frac{2}{N} < x$. For $n \geq N$ we have $\frac{2}{n} \leq \frac{2}{N} < x$, and so $f_n(x) = 0$.

(b) Suppose towards a contradiction that f_n converges uniformly to some function $g : [0, 1] \rightarrow \mathbb{R}$. Then f_n also converges pointwise to g , and so by part (a) we have $g(x) = 0$ for all $x \in [0, 1]$.

Consider $\epsilon = 1$. As f_n converges uniformly to g , there must be some N so that

$$|f_n(x) - g(x)| < 1 \quad \text{for all } n \geq N \text{ and } x \in [0, 1].$$

However, for $x = \frac{1}{n}$ we have

$$|f_n(\frac{1}{n}) - g(\frac{1}{n})| = |1 - 0| = 1$$

for any $n \geq N$, which is a contradiction. □

Problem 2. Consider the series of functions $\sum_{n=0}^{\infty} \frac{x^n}{1+x^n}$.

- (a) Prove that for any $r \in (0, 1)$, the series converges uniformly on $[0, r]$.
- (b) Prove that the series does not converge uniformly on $[0, 1)$.

Solution: (a) Fix $r \in (0, 1)$. Then for $x \in [0, r]$ we have

$$\left| \frac{x^n}{1+x^n} \right| \leq \frac{r^n}{1+0} = r^n.$$

Note that the geometric series $\sum r^n$ converges since $r < 1$. Therefore, $\sum \frac{x^n}{1+x^n}$ converges uniformly on $[0, r]$ by the Weierstrass M -test.

(b) Let $s_n(x) = \sum_{k=0}^n \frac{x^k}{1+x^k}$ denote the n th partial sum. Note that $s_n \in C([0, 1))$, since each function $\frac{x^k}{1+x^k}$ is continuous on the larger interval $[0, 1]$, and thus is continuous and bounded on $[0, 1)$ by the extreme value theorem. On the other hand,

$$d_{\text{sup}}(s_n, s_{n-1}) = \sup_{x \in [0, 1)} \left| \frac{x^n}{1+x^n} \right| = \frac{1}{2}$$

for any $n \geq 1$. (In the last equality, we noted that the function $f(x) = \frac{x^n}{1+x^n}$ is increasing on $[0, 1)$ since $f'(x) = nx^{n-1}(1+x^n)^{-2} \geq 0$ for $x \geq 0$, and so the supremum of f must occur at $x = 1$.) Therefore, the sequence $\{s_n\}_{n \geq 0}$ is not Cauchy in $C([0, 1))$, and so cannot be converging uniformly. $\square$

Problem 3. Consider the power series $\sum_{n=1}^{\infty} \frac{x^n}{\sqrt{n}}$.

- (a) Find its radius of convergence R .
- (b) Determine if the series converges at each of the endpoints $x = R$ and $x = -R$.

Solution: (a) Let $c_n = \frac{1}{\sqrt{n}}$. Then

$$|c_n|^{1/n} = \frac{1}{\sqrt[n]{n}} \longrightarrow 1$$

as $n \rightarrow \infty$ (by the limit law for division, HW4#5, and Lecture 15). Therefore $\limsup |c_n|^{1/n} = 1$, and so $R = 1$.

(b) At $x = 1$, we have the series $\sum \frac{1}{\sqrt{n}}$. This diverges by Lecture 16.

At $x = -1$, we have the series $\sum \frac{(-1)^n}{\sqrt{n}}$. This converges by the alternating series test, since $\frac{1}{\sqrt{n}} \rightarrow 0$ as $n \rightarrow \infty$.

Altogether, we conclude that the power series converges if and only if $x \in [-1, 1)$. $\square$

Problem 4. For each of the following power series, determine the radius of convergence.

$$(a) \sum_{n=0}^{\infty} n^2 x^n \qquad (b) \sum_{n=1}^{\infty} \left(\frac{x}{n}\right)^n \qquad (c) \sum_{n=0}^{\infty} n! x^n$$

Solution: (a) Let $c_n = n^2$. Then

$$|c_n|^{1/n} = (n^{1/n})^2 \longrightarrow 1$$

as $n \rightarrow \infty$ (by the limit law for multiplication and Lecture 15). Therefore $\limsup |c_n|^{1/n} = 1$, and so $R = 1$.

(b) Let $c_n = n^{-n}$. Then

$$|c_n|^{1/n} = \frac{1}{n} \longrightarrow 0$$

as $n \rightarrow \infty$. Therefore $\limsup |c_n|^{1/n} = 0$, and so $R = \infty$.

(c) Let $c_n = n!$. Then

$$|c_n|^{1/n} = (n!)^{1/n} \longrightarrow \infty$$

as $n \rightarrow \infty$ (by Lecture 15). Therefore $\limsup |c_n|^{1/n} = \infty$, and so $R = 0$. □

Problem 5. The *hyperbolic sine* function is defined by $\sinh x = \frac{1}{2}(e^x - e^{-x})$.

- (a) Find its Taylor series centered at $x = 0$.
- (b) Prove that this series converges to $\sinh x$ for any $x \in \mathbb{R}$.

Solution: (a) Set $f(x) = \sinh x$. As $\frac{d}{dx}e^x = e^x$, we have $\frac{d}{dx}e^{-x} = -e^{-x}$. Therefore

$$f^{(n)}(x) = \begin{cases} \frac{1}{2}(e^x - e^{-x}) & \text{if } n \text{ is even,} \\ \frac{1}{2}(e^x + e^{-x}) & \text{if } n \text{ is odd,} \end{cases}$$

and so

$$f^{(n)}(0) = \begin{cases} 0 & \text{if } n \text{ is even,} \\ 1 & \text{if } n \text{ is odd.} \end{cases}$$

So the Taylor series of $\sinh x$ centered at $x = 0$ is

$$\sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}.$$

- (b) Fix $x_0 \in \mathbb{R}$. Set $r = |x_0| + 1$, so that $x_0 \in (-r, r)$. Then

$$|f^{(n)}(x)| = \left| \frac{1}{2}(e^x - (-1)^n e^{-x}) \right| \leq \frac{1}{2}(e^x + e^{-x}) \leq \frac{1}{2}(e^r + e^r) = e^r \quad \text{for all } x \in (-r, r),$$

for any $n \geq 0$. Therefore, by Lecture 36, we have

$$\sinh x = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}$$

for any $x \in (-r, r)$. In particular, equality holds for the point $x = x_0$ we started with. As $x_0 \in \mathbb{R}$ was arbitrary, we conclude that equality is true for any $x_0 \in \mathbb{R}$. $\square$

Problem 6. Consider the function $f : (-1, \infty) \rightarrow \mathbb{R}$, $f(x) = \log(1+x)$.

- (a) Using induction, prove that $f^{(n)}(x) = (-1)^{n+1}(n-1)!(1+x)^{-n}$ for all $n \geq 1$.
- (b) Using part (a), find the Taylor series of f centered at $x = 0$.
- (c) Suppose we want to approximate the number $f(1) = \log 2$. Prove that for any $n \geq 2$,

$$\left| f(1) - \sum_{k=1}^{n-1} \frac{(-1)^{k+1}}{k} \right| \leq \frac{1}{n}.$$

This gives us a way to approximate $\log 2$ and to guarantee that the approximation is as accurate as we want, without having to know the value of $\log 2$ beforehand.

Solution: (a) Base case: For $n = 1$, we have $f'(x) = (1+x)^{-1}$ by the chain rule.

Inductive hypothesis: Assume that

$$f^{(n)}(x) = \frac{(-1)^{n+1}(n-1)!}{(1+x)^n}$$

for some $n \geq 1$. Then, by the chain rule we have

$$f^{(n+1)}(x) = (-n) \frac{(-1)^{n+1}(n-1)!}{(1+x)^{n+1}} = \frac{(-1)^{n+2} n!}{(1+x)^{n+1}},$$

and so the statement is true for $n+1$.

Therefore, by induction, we conclude that the formula holds for all $n \in \mathbb{N}$.

(b) By part (a),

$$\frac{f^{(n)}(0)}{n!} = \frac{(-1)^{n+1}(n-1)!}{n!} = \frac{(-1)^{n+1}}{n}$$

for any $n \geq 1$. When $n = 0$, we have $f(0) = 0$. Therefore the Taylor series of f centered at $x = 0$ is

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n.$$

(c) Fix $n \geq 2$. By Taylor's theorem with remainder, there exists a point $y \in (0, 1)$ so that

$$f(1) = \sum_{k=1}^{n-1} \frac{(-1)^{k+1}}{k} (1-0)^k + \frac{f^{(n)}(y)}{n!} (1-0)^n.$$

Rearranging this expression and using part (a), we find

$$\left| f(1) - \sum_{k=1}^{n-1} \frac{(-1)^{k+1}}{k} \right| = \frac{|f^{(n)}(y)|}{n!} = \frac{1}{n|1+y|^n} \leq \frac{1}{n}.$$

In the last step, we noted that $y > 0$. □